



Systemic light chain and transthyretin amyloidosis—treatment advancements and future directions

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Once rapidly fatal, neglected, and orphan diseases without approved therapeutic options, systemic amyloidoses are now highly treatable. Basic scientific discoveries regarding the molecular mechanisms of transthyretin misfolding and aggregation have driven the development of drugs for the treatment of transthyretin (ATTR) amyloidosis, with six novel therapies approved since 2018. The combination of daratumumab, cyclophosphamide, bortezomib, and dexamethasone was approved for the treatment of light chain (AL) amyloidosis in 2021. Since then, highly effective immunotherapies, such as bispecific T-cell engagers, have been tested in clinical trials in patients with relapsed AL amyloidosis, with unprecedented efficacy. Thus, suspicion of systemic amyloidosis should be raised early in individuals with a suggestive clinical presentation, followed by rapid tissue diagnosis, with appropriate protein typing and prompt commencement of active therapy. These rapid and exciting therapeutic advancements set the stage for this Review on the epidemiology, pathophysiology, and diagnosis of systemic amyloidosis. Through the lens of the molecular mechanisms underlying amyloidosis pathogenesis, we will focus on approved and investigational therapies. We also discuss future directions and challenges concerning a cure for patients with AL or ATTR amyloidosis.

Introduction

Amyloidoses are a family of diseases with a common root: organised deposition of multimers of a conformationally unstable protein in target organs.¹ The term amyloidosis epitomises the mysterious nature of this disease. From the ancient Greek *ἄμυλον*, in Latin *amylum*, Rudolph Virchow borrowed this term from botanist Matthias Schleiden, who had used it to describe a starch component of plants.² Similar to starch, Virchow noted that amyloid deposits stained bright red-purple with iodine sulphuric acid.³ Previously mistaken as a polysaccharide or lipid due to its waxy appearance, the structure of amyloid as non-branching protein fibrils, observed with electron microscopy, was finally reported in 1959 by Cohen and Calkins.⁴ In 1971, Glenner and colleagues proved that immunoglobulin free light chains (FLCs) could form amyloid.⁵

Since these pioneering observations, remarkable progress has been made in understanding the aetiology and pathogenesis of these complex disorders. Building on fundamental basic scientific discoveries, therapies targeting the production or stability of amyloidogenic proteins entered clinical development in the late 2010s, leading to the historic approval of the first treatments for both light chain (AL) amyloidosis and transthyretin (ATTR) amyloidosis in recent years.

Given these extraordinary advances, the state-of-the-art therapeutic approach to patients diagnosed with AL and ATTR amyloidosis should be reviewed. Although a common pathogenic mechanism underlies all types of amyloidoses, their aetiology and clinical presentation are rather distinct. We discuss these points in subsequent sections.

Epidemiology

Amyloidoses can be classified as either localised or systemic. In systemic amyloidosis, amyloid protein

synthesis occurs at an anatomically distant site from its target organ (appendix p 1).⁶ Amyloidoses can be inherited, with an autosomal-dominant pattern, or acquired. More than 40 amyloid proteins have been identified in humans, and 14 can cause systemic amyloidosis, with immunoglobulin light chain and transthyretin causing approximately 95% of cases.⁷ Amyloid subtypes are classified according to the amyloid precursor protein, abbreviated as the letter A, followed by a suffix (ie, immunoglobulin light chain is written as AL and transthyretin as ATTR). The protein can be specified further—ie, ATTRv for variant and ATTRwt for wild-type.

Substantial heterogeneity exists in amyloidosis epidemiology and clinical outcomes across genders, races, and ethnicities. Both AL and, particularly, ATTRwt amyloid cardiomyopathies are predominant in men, for reasons that are only partially understood. A protective role for oestrogens has been suggested; however, unbiased screening via nuclear imaging or autopsy showed similar incidences of ATTRwt amyloidosis between men and women, in keeping with underdiagnosis in women.^{8–12} The prevalence of admission to hospital due to amyloidosis in the USA was higher in patients who are non-Hispanic White than in patients who are non-Hispanic Black or Hispanic. Patients who are White were also reported to be older at diagnosis. Data show that ATTRwt amyloidosis is more

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See Online for appendix

Search strategy and selection criteria

Papers cited in this work were searched for on PubMed from database inception to Dec 15, 2025, with appropriate key terms (ie, "transthyretin" AND "cleavage" AND "pathogenesis"). Only peer-reviewed studies in English were included.

frequent in individuals who are non-Hispanic Black, older, and male gender.¹³ In AL amyloidosis, patients who are non-Hispanic Black were reported to have a more aggressive disease phenotype.¹⁴

Although systemic amyloidosis is considered rare across the general population, it can be identified frequently in several specific clinical scenarios.¹⁵ A population-based study reported an AL amyloidosis incidence of 12 per 1000000 people per year, with an estimated prevalence of 30000–45000 cases in Europe and the USA.¹⁶ ATTRv amyloidosis presents clusters of specific mutations with highly variable penetrance in some protein locations, such

as the p.Val50Met variant in Portugal, Sweden, Japan, Brazil, Cyprus, and Majorca, and the p.Thr80Ala variant—endemic in County Donegal, Ireland. The p.Val142Ile variant is the most common in the USA, with a reported prevalence of 3·5% in the Black population.¹⁷ ATTRwt cardiomyopathy was detected in 2·5–14% of patients with heart failure with preserved ejection fraction via systematic, unbiased screening or tissue biopsy.^{9,18,19} However, when risk factors are accounted for, such as pre-existing bilateral carpal tunnel syndrome in older men without obesity, these values can approach 20%.²⁰ The true prevalence of both ATTRv and ATTRwt amyloidosis is

	AL	ATTRwt	ATTRv	Tests
Cardiovascular				
Congestive HF (dyspnoea, pleural effusions, leg oedema, ascites, etc)	X	X	X	ECG; TTE with strain*; NT-proBNP or BNP; troponin T; lipid panel; cardiac MRI; Holter monitoring; cardiac scintigraphy with DPD or PYP†; right heart catheterisation with RV endomyocardial biopsies
Arrhythmia (AF or atrial flutter, VT)	X	X	X	As above
Fatigue and exertional dyspnoea	X	X	X	As above
OH, syncope‡	X	X	X	As above
Progressive decline in blood pressure, or need to taper off anti-hypertensive medications over time‡	X	X	X	As above
Inability to tolerate standard HF therapies (β blockers, ACE inhibitors, ARBs, ARNIs)	X	X	X	As above
Sudden onset, often severe hyperlipidaemia§	X	..	..	As above
Stroke or cardiovascular ischemic events (TIAs, NSTEMI)	X	X	X	As above
Nervous				
Dysaesthesia, allodynia (small fibre neuropathy)	X	..	X	EMG; autonomic tests
Autonomic dysfunction (ie, OH)	X	..	X	As above
Gastrointestinal				
Unintentional weight loss	X	..	..	Liver function tests, including alkaline phosphatase and bilirubin; gastrointestinal endoscopies and biopsies; liver fibroscan
Taste change or loss of taste	X	..	..	As above
Early satiety‡	X	..	..	As above
Dyspepsia, nausea, emesis	X	..	..	As above
Severe, persistent diarrhoea¶	X	..	X	As above plus exocrine pancreatic enzyme deficiency testing
Hepatomegaly without imaging defects	X	..	..	As above
Muscular-skeleton				
Bilateral carpal tunnel syndrome	X	X	X	EMG; spinal MRI
Spinal stenosis, pseudo-claudication	..	X	X	As above
Bicep tendon rupture	..	X	..	As above
Other tendinopathies	..	X	..	As above
Kidney				
Proteinuria, often in the nephrotic range	X	..	..	Complete metabolic panel; urine protein–creatinine ratio; urine albumin–creatinine ratio; UPEP plus IFE
Peripheral oedema, ascites, anasarca	X	..	..	As above
Progressive renal failure	X	..	X	As above
Skin and soft tissue				
Purpura on the face or neck	X	..	..	CBC, PT, or INR, aPTT, Factor X concentration, monoclonal protein studies (SPEP plus IFE, UPEP plus IFE, and serum FLC)
Macroglossia	X	..	..	As above
Muscle pseudo-hypertrophy (shoulder pad)	X	..	..	As above
Hoarseness	X	..	..	As above

(Table 1 continues on next page)

	AL	ATTRwt	ATTRv	Tests
(Continued from previous page)				
Haematology				
Anaemia	X	..	..	aPTT; PT or INR; factor X concentration; iron studies; monoclonal protein studies (SPEP plus IFE, UPEP plus IFE, and serum FLC); bone marrow aspirate or biopsy for morphological review; flow cytometry analysis; FISH for plasma cell disorder abnormalities; Congo red stain
Other cytopenia	X	..	..	As above
Venous thrombosis	X	..	..	As above
Bleeding or easy bruising	X	..	..	As above
Endocrine				
Hypothyroidism§	X	..	..	TSH and free thyroxine
Genitourinary				
Erectile dysfunction, vaginal dryness	X	X	X	Free and total testosterone

ACE=angiotensin converting enzyme. AF=atrial fibrillation. AL=light chain. aPTT=activated prothrombin time. ARB=angiotensin receptor blocker. ARNI=angiotensin receptor neprilysin inhibitor. ATTR=transthyretin amyloidosis. ATTRv=transthyretin amyloidosis variant. ATTRwt=transthyretin amyloidosis wild-type. CBC=cell blood count. DPD=3,3'-diphosphono-1,2-propanodicarboxylic acid. ECG=electrocardiography. EMG=electromyography. FISH=fluorescence in situ hybridisation. FLC=free light chains. HF=heart failure. IFE=immunofixation. INR=international normalised ratio. NSTEMI=non-ST-segment elevation myocardial infarction. OH=orthostatic hypotension. PT=prothrombin time. PYP=^{99m}Tc-labelled pyrophosphate. RV=right ventricle. SPEP=serum protein electrophoresis. TIAs=transient ischaemic attacks. TSH=thyroid stimulating hormone. TTE=transthoracic echocardiography. UPEP=urine protein electrophoresis. VT=ventricular tachycardia. *Cardiac MRI is more sensitive and specific than TTE and should be considered before excluding amyloidosis based on TTE. †For patients in whom a plasma cell disorder has been ruled out and ATTR is suspected. ‡Typically secondary to autonomic dysfunction. §Typically secondary to nephrotic syndrome. ¶Diarrhoea can be secondary to autonomic dysfunction or, more rarely, pancreatic insufficiency. Constipation can also occur secondary to autonomic dysfunction. ||A 24-h urine collection is key to accurately ascertain proteinuria and the presence or absence of a monoclonal spike.

Table 1: Common symptoms, signs, and laboratory or imaging findings suggesting AL, ATTRwt, or ATTRv amyloidosis, and testing (clinical red flags)

unknown. A prospective study showed an ATTR cardiomyopathy prevalence of 6·8% in older patients with heart failure who are Black or Hispanic.²¹ The availability of effective therapies, combined with the implementation of non-invasive diagnostic techniques, has led to an exponential increase in ATTR cardiomyopathy detection, including early stages.²²

Diagnostic considerations

Systemic amyloidoses are relentless conditions that lead to progressive end-organ damage and failure. Heart involvement dictates survival, and early diagnosis is essential to allow for intervention before irreversible cardiac dysfunction occurs. Clinical and laboratory red flags for amyloidosis are reported in table 1, figure 1, and the appendix (p 2). Although the individual sensitivity of each red flag is low, clustering of several red flags should warrant investigation for suspected systemic amyloidosis. An in-depth medical history, including family history, and an accurate physical examination are pivotal. However, pathognomonic manifestations, such as macroglossia or periorbital purpura, are frequently associated with advanced amyloid cardiomyopathy. Therefore, use of biochemical and imaging markers is important to aid in early detection.

An increased serum concentration of monoclonal FLC is detectable in patients with AL amyloidosis years before diagnosis.²³ Screening with markers of amyloid organ involvement, namely NT-proBNP for the heart, albuminuria for the kidney, and alkaline phosphatase for the liver, can detect AL amyloidosis in approximately

1% of asymptomatic patients with an underlying plasma cell disorder.^{24,25} We recommend checking these markers at least yearly in patients who are asymptomatic with monoclonal gammopathy of undetermined significance or smouldering multiple myeloma and elevated FLC.²⁶ Serum testing for amyloidogenic λ FLC, based on the protein kinetic instability, has shown good sensitivity and specificity, and might soon enter clinical use.²⁷

Screening efforts to identify patients with monoclonal gammopathy of undetermined significance or smouldering multiple myeloma harbouring an amyloid-biased immunoglobulin light chain (IgL) variable region genotype are underway.²⁸ Machine learning algorithms to predict the amyloidogenic potential of IgL according to its amino acid sequence have been developed and might facilitate early diagnosis.²⁹

Carpal tunnel syndrome, particularly if bilateral, is disproportionately prevalent in patients with ATTRwt or ATTRv and typically precedes cardiac manifestations by 5–10 years.³⁰ Prospective studies are required to understand the impact, timing of initiation, and frequency of cardiac scintigraphy screening for ATTR cardiomyopathy in patients with a history of bilateral carpal tunnel syndrome.

Conduction abnormalities, poor R-wave progression, pseudo-infarct pattern, or atrial fibrillation or flutter on electrocardiogram, although non-specific, are clues to a diagnosis of AL amyloidosis. Low-voltage QRS complexes in limb leads are common in advanced amyloid cardiomyopathy and are a predictor of mortality: they do not aid early diagnosis.³¹ Artificial intelligence

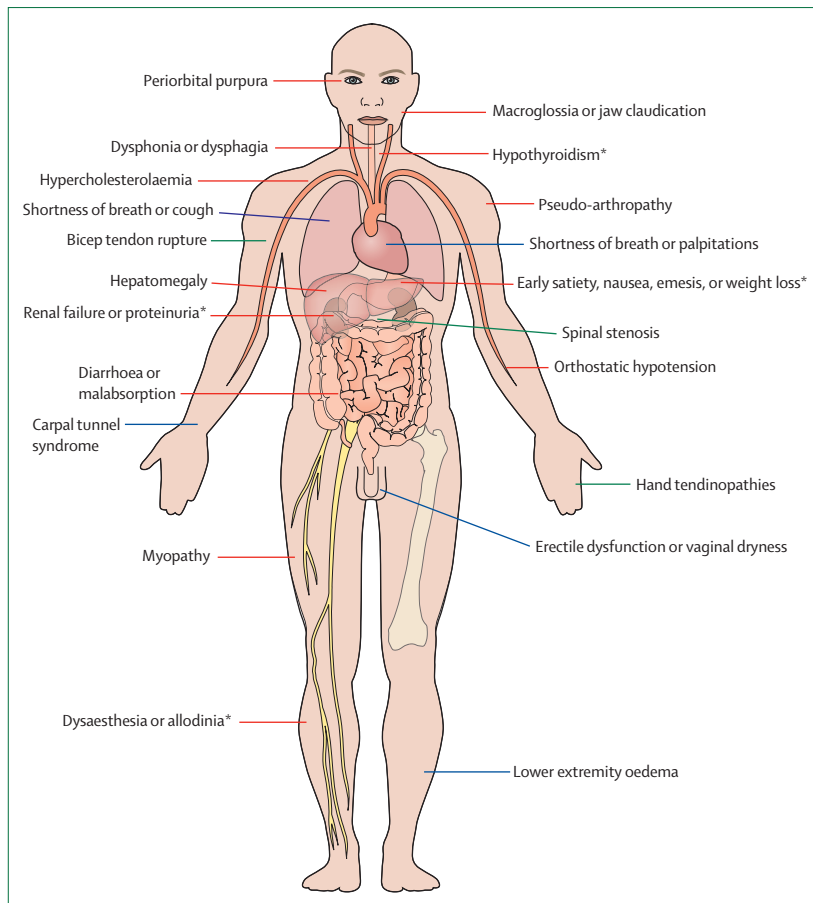


Figure 1: Potential organs or systems involved by AL or ATTR amyloidosis

Red colour indicates organs or systems mostly or uniquely involved by AL amyloidosis. Green colour indicates those mostly or uniquely involved by ATTRwt amyloidosis. Blue colour indicates sites of clinical presentation common to both. AL=light chain amyloidosis. ATTR=transthyretin amyloidosis. ATTRv=transthyretin amyloidosis variant. ATTRwt=transthyretin amyloidosis wild-type. *Also common in ATTRv.

has been successfully applied to electrocardiography and transthoracic echocardiography (TTE) for the early detection of cardiac amyloidosis based on initial findings (table 1, appendix p 2).^{32,33} However, these algorithms cannot currently differentiate between AL and ATTR cardiomyopathies.^{34–36} When done by trained sonographers, TTE is a key tool for recognising cardiac amyloidosis. Concentric thickening of the ventricular walls, atrial enlargement, and reduced global longitudinal strain with apical sparing, along with signs of diastolic failure, are suspicious for amyloid cardiomyopathy.³⁷

Compared with TTE, cardiac magnetic resonance provides more granular information on myocardial tissue composition. In patients with cardiac amyloidosis, the extracellular volume is typically increased, reflecting the deposition of amyloid in the extracellular space between cardiomyocytes.³⁸

For patients in whom ATTR cardiomyopathy is suspected and screening for a plasma cell disorder is

negative, a non-invasive diagnosis can be reached with a bone-avid tracer (^{99m}Tc-labelled pyrophosphate, 3-diphosphono-1,2-propanodicarboxylic acid, or hydroxymethylene diphosphonate) cardiac SPECT (figure 2).²² However, specific variants (eg, p.Phe84Leu, p.Ser97Tyr, and, less often, p.Val50Met) do not show uptake of bone tracers and false negative results are expected, outlining the importance of concomitant transthyretin (TTR) gene sequencing.^{42,43}

PET tracers developed for cerebral amyloid detection in Alzheimer's disease bind to various types of amyloid in the heart, with distinct uptake patterns between AL and ATTR cardiomyopathies.^{44,45} Since 2023, the pan-amyloid tracer ¹²⁴I-evuzamitide (¹²⁴I-p5+14, AT-01) has been shown to accurately and sensitively detect cardiac amyloid deposition in patients with AL, ATTRwt, and other rare forms of amyloidosis.^{45,46} Costs and technical limitations are the main barriers to the widespread use of these imaging techniques.⁴⁷

Once a clinical suspicion of systemic amyloidosis is raised, the presence of amyloid should be substantiated by tissue analysis, unless diagnostic criteria for non-invasive ATTR cardiomyopathy are met (figure 2, appendix pp 3–4). Amyloid typing is required because distinct types of amyloidosis have overlapping demography and clinical presentation. Liquid chromatography–mass spectrometry is the gold standard, enabling sensitive and specific identification of the amyloid precursor protein in an unbiased manner in Congo red-positive tissue. However, accessibility and costs might be prohibitive.^{48–50} Immunohistochemistry and immunofluorescence are sensitive and specific when done in large referral centres. The major caveats are masking or loss of antigen due to tissue fixation or inherent to the process of fibril formation itself (ie, proteolytic cleavage of antigenic determinants), marked background staining, and limited availability of validated antibodies for the identification of distinct subtypes.⁵¹ The need for proper instrumentation and technical expertise hampers the widespread use of electron microscopy with immunogold, which couples the specificity of antibodies to the resolution of electron microscopy.⁵² Once a diagnosis of amyloid type is firmly established, organ involvement can be determined with appropriate testing (table 1).

Molecular mechanisms underlying amyloidogenesis

All amyloid fibrils share a cross- β -sheet chemical structure characterised by theoretically indefinite repeats of β -sheet motifs.⁵³ A genome-wide analysis revealed that nearly any protein can form β -pleated sheets, yet only a few proteins cause pathogenic amyloid deposition.⁵⁴ These data support a crucial role for factors other than amino acid sequence in determining amyloid formation.

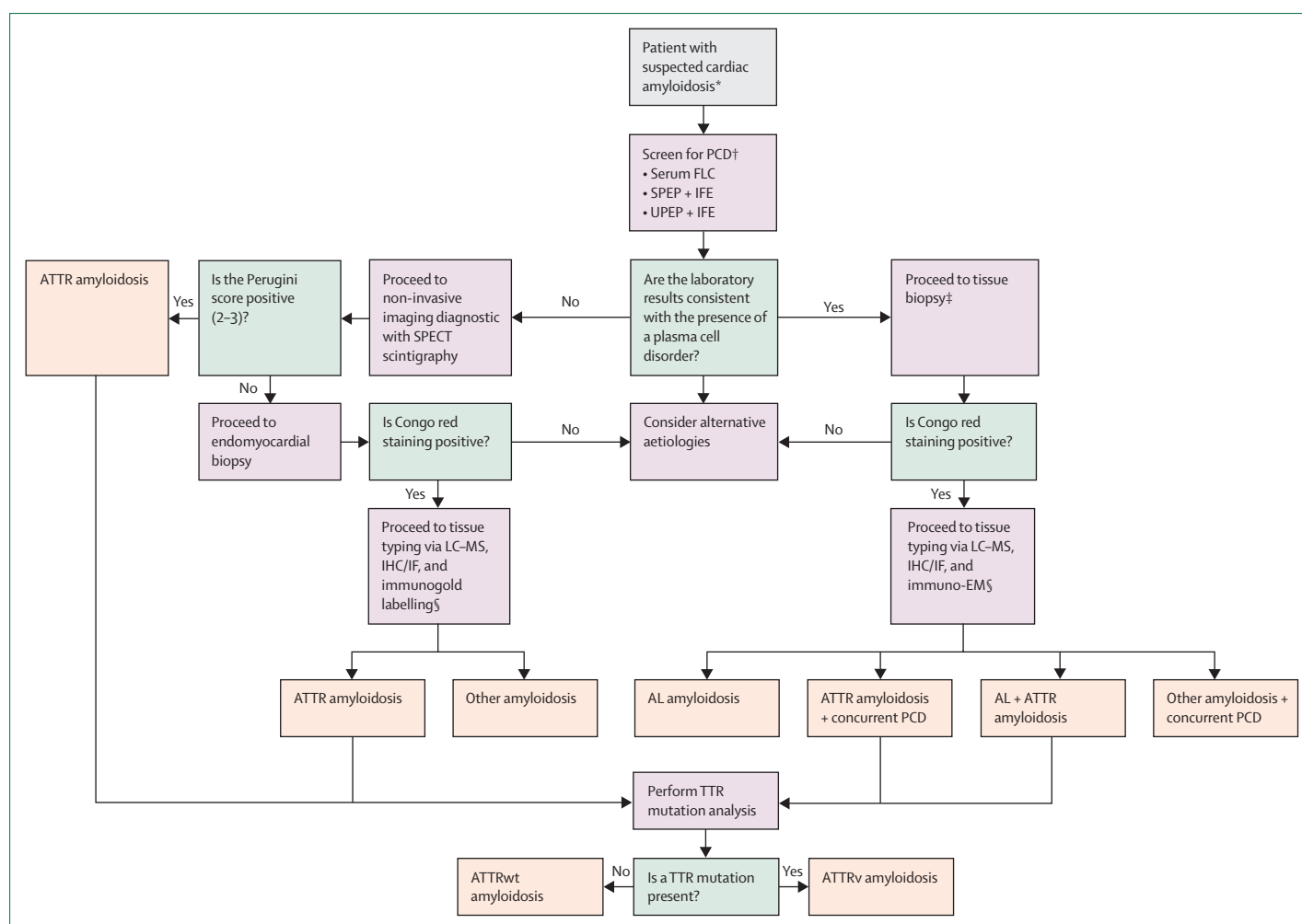


Figure 2: Diagnostic flowchart for patients with suspected amyloid cardiomyopathy

Initially, evaluate for the presence of a plasma cell disorder by doing serum and urine immunofixation electrophoresis and serum FLC measurement and ratio.³⁹ In the presence of a monoclonal protein, a biopsy documenting the amyloid deposits is crucial because clinically, AL and ATTR amyloidosis cannot be reliably distinguished, and the high prevalence of monoclonal proteins (23%)⁴⁰ in patients with ATTR amyloidosis is an important confounding factor. Rare cases of two types of amyloidosis in the same patient have been reported.⁴¹ AL=light chain amyloidosis. ATTR=transthyretin amyloidosis. EM=electron microscopy. FLC=free light chains. IF=immunofluorescence. IFE=immunofixation. IHC=immunohistochemistry. LC-MS=liquid chromatography-mass spectrometry. PCD=plasma cell disorder. SPEP=serum protein electrophoresis. TTR=transthyretin. UPEP=urine protein electrophoresis. *Refer to table 1, appendix (p 2), and figure 1 for signs, symptoms, and test findings raising concern for systemic amyloidosis. †Early consultation with a specialist in PCDs is strongly recommended. ‡See appendix (pp 3–4) for algorithm. §LC-MS is the gold standard for amyloid typing. If not available, IF, IHC, or immuno-EM can be used instead, taking into account masking or loss of antigen due to tissue fixation or inherent to the process of fibril formation itself, marked background staining, and limited availability of validated antibodies.

Amyloid precursor protein intrinsic factors

The amino acid sequence is a main determinant of protein aggregation by modulating conformational stability. For example, most TTR mutations push the equilibrium between circulating, stable TTR tetramers and conformationally unstable monomers towards the monomeric form, facilitating amyloid formation.⁵⁵ However, disease penetrance is highly variable. Gelsolin and the APOA-I Leu75Pro variant have a disease penetrance approaching 100%.^{56,57} Vice versa, p.Val142Ile TTR has a 39% cardiomyopathy phenotypic penetrance in patients aged 60 years or older.²¹ For unclear reasons, the cumulative clinical penetrance of p.Val50Met is 50% in people from France, but 91% in individuals from

Portugal by age 70 years, with the caveat that late-onset manifestations and premature mortality might bias observed frequencies.^{58,59} AL amyloidosis tends towards the λ subtype, with specific rearrangements of the variable region occurring at a disproportionately high frequency.^{60,61} Furthermore, growing evidence suggests that glycosylation and somatic hypermutation might enhance FLC aggregation.^{62,63}

The concept of seeding is common to all amyloidosis. The model posits that a nucleus of misfolded protein aggregates forms over time, serving as a seed that catalyses amyloid deposition at an exponential rate.⁶⁴ This model explains the variably long, asymptomatic phase of systemic amyloidosis, followed by rapidly

progressive symptoms and signs, and provides the rationale for early treatment in patients at risk. A substantial and persistent increase in the concentration of an otherwise normal protein can also promote aggregation via aberrant glycation or proteolysis, as observed for β 2-microglobulin and amyloid A, respectively.^{65–67}

Amyloid precursor protein extrinsic factors

The variable penetrance, latency of clinical symptoms, and heterogeneous severity of the phenotype and the pattern of organ deposition suggest that protein-extrinsic factors contribute to amyloidosis pathogenesis. Extracellular chaperones are key for maintaining proteostasis by interacting with unstable proteins, thereby mitigating the risk of aggregation.⁶⁸ For example, clusterin co-localises with both AL and ATTRwt or ATTRv fibrils, presumably to

maintain their solubility.^{69,70} Interestingly, TTR itself is an extracellular chaperone for β -amyloid peptide.⁷¹

Intracellular and extracellular proteostasis are closely intertwined. Secreted proteins are handled by the endoplasmic reticulum quality control (ERQC), a protein network consisting of chaperones, enzymes, and degradation machinery.⁷² The ERQC determines the fate of nascent polypeptides along the secretory pathway based on whether they are structurally stable for secretion or should be degraded. Evidently, amyloidogenic proteins are sufficiently structurally stable to evade ERQC and are secreted. The decline in proteostasis associated with ageing contributes to the exocytosis of unstable proteins, thereby increasing the risk of protein aggregation disorders in older individuals.^{73,74} Pharmacological modulation of the ERQC was shown to decrease secretion of amyloidogenic

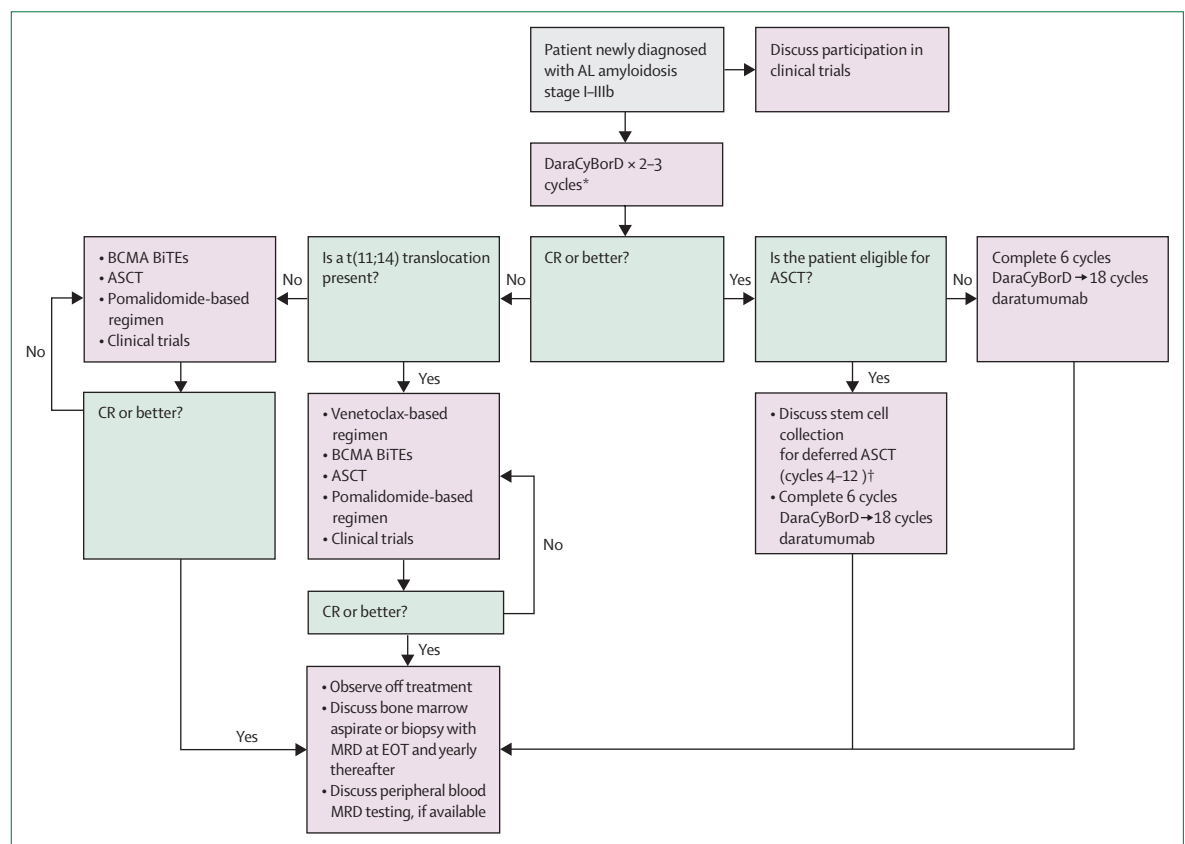


Figure 3: Treatment approach for patients who are newly diagnosed with AL amyloidosis

We encourage discussion of participation in clinical trials with every patient. Our approach is to typically use DaraCyBorD for all patients, regardless of staging and transplantation eligibility, unless there are obvious patient-related contraindications or preferences. The goal of therapy is to reach complete haematological remission by cycle two. If this does not occur, the next treatment approach is determined by the presence of a t(11;14) translocation. If present, a venetoclax-based regimen is preferred, if the drug can be secured. If not present or venetoclax is not available, we recommend considering stem-cell transplantation in patients eligible, or BCMA-targeting agents, or pomalidomide-based treatment. For patients eligible for stem-cell transplantation that have complete haematological remission by cycle two to four with DaraCyBorD, we recommend stem-cell collection between cycles four and 12, followed by completion of DaraCyBorD, as per the Andromeda protocol.⁹⁷ If possible, testing for MRD in blood or bone marrow should be done regularly. We do not recommend maintenance in patients with AL amyloidosis in a complete response. AL=light chain amyloidosis. ASCT=autologous stem-cell transplantation. BCMA=B-cell maturation antigen. BiTEs=bispecific T-cell engagers. CR=complete haematological remission. DaraCyBorD: daratumumab–cyclophosphamide–bortezomib–dexamethasone. EOT=end of treatment. MRD=minimal residual disease. *We recommend aiming for complete haematological remission by cycle two. If this goal is not reached, we recommend changing therapy by cycle three to maximise the chances of organ function preservation and survival. †Candidacy for ASCT should be reassessed regularly, because it is a dynamic assessment based on patient-related factors that might change over time.

proteins, providing the basis for rational design of novel therapies.^{75,76}

Proteomic studies identified amyloid-associated proteins shared across all subtypes, such as serum amyloid P, and others specific to distinct types of amyloidosis.⁷⁷ Preclinical studies showed some of these cofactors to be pathogenic by facilitating aggregation or inhibiting phagocytosis of fibrils.^{78–81}

The scarcity of clinically relevant animal models that reproduce amyloidosis hampers understanding of the microenvironment's pathogenic role in amyloid deposition.⁸² Macrophages and neutrophils are the predominant phagocytic cells—preliminary evidence suggests a quantitative or qualitative deficiency in these cell populations in amyloid-involved organs.^{83–87} In vitro, amyloid fibrils increase the proliferation and upregulation of inflammatory transcriptional signatures in fibroblasts, suggesting a causative role in driving fibrosis and irreversible organ damage.^{88–90}

Comprehensive strategies for treating patients with AL amyloidosis

The treatment approach to AL amyloidosis involves three separate, yet complementary, aspects: (1) prompt and sustained FLC reduction to target pathogenesis; (2) individualised therapy based on patient-specific (eg, frailty, comorbidities, and preferences) and disease-specific (eg, staging and organ involvement) factors to optimise outcomes; and (3) optimal organ-specific supportive care to minimise iatrogenic complications and maximise quality of life. Improving quality of life includes addressing organ-specific needs, managing symptoms, and offering psychosocial support to patients and families throughout the treatment journey. Proper AL amyloidosis

staging is important to guide therapeutic decisions and counselling (appendix pp 5–6).^{91–94} As the treatment landscape of AL amyloidosis evolves, existing staging systems might lose their discriminating power, as seen with the Mayo 2012 staging system in patients receiving daratumumab–cyclophosphamide–bortezomib–dexamethasone (DaraCyBorD).⁹⁵

Treatment approaches for patients who are newly diagnosed

Highly effective plasma cell-directed therapies with high rates of haematological and organ responses have led to an improvement in outcomes for patients with AL amyloidosis (appendix pp 7–8).⁹⁶ DaraCyBorD is the first and only licensed treatment for patients with stage I–IIIA AL amyloidosis.⁹⁷ The International Society of Amyloidosis Working Group issued comprehensive guidelines on stem-cell transplant and non-transplant approaches for the treatment of patients with AL amyloidosis.^{98,99} Our consensus approach is shown in figure 3, with the sites of activity of distinct therapeutic agents outlined in figure 4.

High-dose intravenous melphalan, followed by peripheral blood autologous stem-cell transplantation, a cornerstone of treatment for select patients with AL amyloidosis since the mid-1990s, leads to deep and long-term remissions, with approximately 40% of patients having a haematological complete response lasting an average of 12·3 years.¹⁰⁰ However, due to various factors, such as poor performance status, advanced organ dysfunction, and inadequate caregiver support, only 10–20% of patients who are newly diagnosed are eligible for autologous stem-cell transplantation.⁹⁸ As novel therapies, such as bispecific

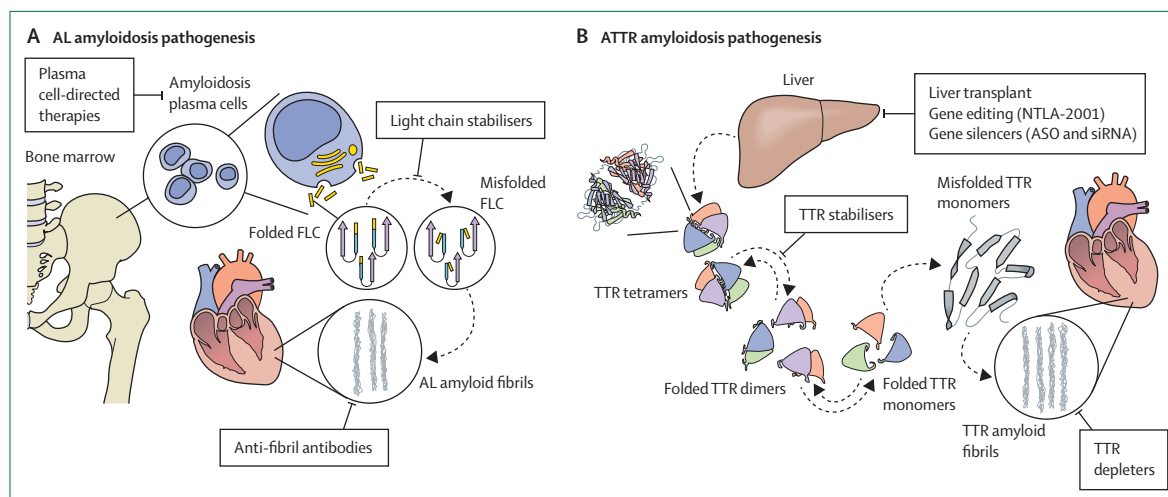


Figure 4: Pathogenesis of AL and ATTR amyloidosis

(A) Pathogenesis of AL amyloidosis from plasma cells expanding in the bone marrow through secretion of folded FLCs, misfolding of circulating FLCs, and finally, formation of AL amyloid fibril deposition in the heart and other organs. (B) Pathogenesis of ATTR amyloidosis from synthesis and secretion of TTR tetramers by the liver, TTR dimer and monomer formation, misfolding of TTR monomers leading to TTR amyloid formation, and finally, deposition in the heart and other organs. The sites of activity of approved and investigational therapies are outlined in rectangles for both disorders. AL=light chain amyloidosis. ASO=antisense oligonucleotides. ATTR=transthyretin amyloidosis. FLC=free light chains. TTR=transthyretin.

T-cell engagers (BiTEs), lead to frequent, rapid, and deep haematological responses with manageable toxicities, the use of autologous stem-cell transplantation is expected to decline over time.^{101–104}

For patients with stage I–IIIb AL amyloidosis, the preferred treatment is DaraCyBorD, which yields high haematological response rates, with a 78% very good partial response or better, and approximately 50–55% organ response at 18 months, resulting in significant improvement in overall survival.⁹⁷ Careful consideration of patient characteristics is paramount. Patients with severe neuropathy who cannot tolerate bortezomib can be treated with daratumumab monotherapy, venetoclax in the presence of a t(11;14) translocation, immunomodulatory drug-based regimens, or BiTEs.¹⁰⁵

Patients with stage IIIb cardiac involvement should be treated cautiously with careful management of volume status to mitigate the risk of decompensated heart failure. Early dose reduction of dexamethasone is often warranted.¹⁰⁶ Daratumumab monotherapy in these patients leads to a haematological overall response rate of 74%, with a 23% complete response and a 29% very good partial response. The median time to very good partial response was 1·8 months (range 0·2–7·3), with a median overall survival of 10·3 months (95% CI 4·1–32·1).¹⁰⁷ Retrospective studies of DaraCyBorD in patients with stage IIIb AL amyloidosis are promising, with an overall response rate of 100% and a very good partial response or better of nearly 90%.^{106,108} Cardiac responses were frequent and rapid, with manageable infectious complications and 67·5% 1-year overall survival. The few patients reported, short duration of follow-up, and retrospective nature of these studies preclude firm conclusions. In the context of large amyloidosis centres with multidisciplinary care, DaraCyBorD should be considered as the preferred treatment strategy in patients with stage IIIb AL amyloidosis.

Treatment approaches for patients who have relapsed

Currently, no consensus exists on the criteria for initiating second-line therapy in patients with progressive disease. Early treatment is crucial because organ progression due to amyloid deposition can occur, even with modest FLC increments.¹⁰⁹ Participation in clinical trials should be considered, as patients can receive promising investigational agents. No approved anti-plasma cell therapies exist in the relapsed setting, and optimal sequencing of regimens depends on factors such as depth and duration of initial response, previous agents, patient frailty, and the presence of end-organ damage or residual adverse events. Re-treatment with first-line regimens can be considered if side-effects were manageable and response was rapid, deep, and sustained. Alternatively, treatment with non-cross-resistant agents is preferred. Herein, we examine therapeutic classes that have been evaluated or are under investigation in relapsed AL amyloidosis.

Proteasome inhibitors

Bortezomib is a first-in-class, reversible, boronic acid proteasome inhibitor.¹¹⁰ It produces durable haematological responses, with promising long-term overall survival, as a monotherapy in patients with relapsed AL amyloidosis; it is a mainstay of front-line treatment.¹¹¹ Peripheral neuropathy is a major drug-related adverse event, and careful monitoring of symptoms during therapy with prompt dose reduction, schedule change, or drug discontinuation is crucial to avoid irreversible nerve damage. Ixazomib is an orally bioavailable boronic acid proteasome inhibitor. In a randomised, phase 3 trial, ixazomib with dexamethasone showed improved efficacy compared with the clinician's choice in patients with relapsed AL amyloidosis who are naive to proteasome inhibitor.^{112,113} Ixazomib-based regimen use is generally limited to patients who prefer oral treatment and a more palliative approach. Carfilzomib is an irreversible epoxyketone proteasome inhibitor. Cardiovascular and renal adverse events have been reported in patients with multiple myeloma and carfilzomib use in patients with pre-existing cardiac and renal morbidity is discouraged, limiting its indication in patients with AL amyloidosis.^{114,115}

CD38-targeting antibodies

Daratumumab and isatuximab are CD38-targeting IgGκ monoclonal antibodies (mAbs) with potent single-agent activity in multiple myeloma and AL amyloidosis.¹¹⁶ In prospective trials, daratumumab monotherapy elicited a 70–90% overall response rate and a 15–41% complete response, with 25–28 months progression-free survival.^{117,118} Preliminary data from prospective trials of daratumumab or isatuximab in combination with pomalidomide and dexamethasone in relapsed AL amyloidosis suggest a high overall response rate and a signal for infections.^{119,120}

Immunomodulatory drugs

Immunomodulatory drugs have moderate single-agent activity in AL amyloidosis. In a pooled analysis of 125 patients enrolled in clinical trials incorporating immunomodulatory drugs between 2004 and 2015, the median overall survival was 4·3 years and the median progression-free survival was 2·1 years (95% CI 1·6–2·6).¹²¹ Immunomodulatory drugs are poorly tolerated in patients with advanced cardiomyopathy, resulting in increased cardiac biomarkers. Therefore, a low starting dose is recommended.¹²² Among immunomodulatory drugs evaluated in AL amyloidosis, pomalidomide is the best tolerated and most effective, followed by lenalidomide and thalidomide. The combination of lenalidomide with bortezomib or ixazomib is effective, but data are sparse.¹²³

Alkylators and promising investigational agents

Although alkylators, particularly melphalan, have been a cornerstone of amyloidosis treatment, their use is now scant with the advent of more effective and less toxic treatments.^{124–126}

	Approval status in ATTR amyloidosis or phase of study	Chemical structure and mechanism of action	Dose and route	Efficacy primary outcomes	Noticeable treatment-related adverse events
TTR stabilisers					
Tafamidis	US FDA approved for ATTRv and ATTRwt CA and PN; EMA approved for ATTRv PN	Benzoxazole derivative of NSAID; binds to thyroxine binding site on TTR, leading to tetramer stabilisation	80 mg orally daily (vyndaqel); 61 mg orally daily (vyndamax)	In ATTR CA, compared with placebo at 30 months: 13% absolute risk reduction in mortality and 32% relative risk reduction in cardiovascular-related admissions to hospital	Minimal; interaction with high-dose rosuvastatin ^{142,148}
Acoramidis	EMA and FDA approved for ATTRv and ATTRwt CA	Thr119Met mimicker; leads to the formation of hydrogen bonds between Ser117 residues in each TTR monomer, causing tetramer stabilisation	800 mg orally twice daily	In ATTR CA, compared with placebo at 30 months: 1.8 win ratio in a 4-step hierarchical analysis of all-cause mortality, cardiovascular admissions to hospital, change in baseline NT-proBNP, and change in baseline 6-min walk test	Minimal ^{149,150}
Diflunisal	Off-label use in ATTRv and ATTRwt CA and PN; ongoing phase 2/3 studies	NSAID; binds to thyroxine binding site on TTR, leading to tetramer stabilisation	250 mg orally twice daily	In ATTR PN, compared with placebo at 2 years: 16.3 point reduction in the rate of progression of neurological impairment (NIS+7 nerve test score) and 6.1 point improvement in QOL (SF-36 physical component)	Cardiovascular thrombotic events; gastrointestinal bleeding; acute kidney injury; fluid retention ¹⁵¹
TTR gene silencers					
Inotersen	FDA and EMA approved for ATTRv PN	2'-O-methoxyethyl-modified phosphorothioate ASO; binds to TTR mRNA in the liver, leading to RNaseH-mediated TTR mRNA degradation	300 mg subcutaneously once weekly; vitamin A 3000 IU daily supplementation	In ATTRv PN, compared with placebo at 15 months: 19.7 point reduction in the progression of neurological impairment (NIS+7 nerve test score) and 11.7 point improvement in QOL (Norfolk QOL-DN score)	Thrombocytopenia (severe and sudden in 3% of patients); glomerulonephritis (3% of patients) ¹⁵²
Eplontersen	FDA and EMA approved for ATTRv PN; phase 3 trial in ATTR CA ongoing	GalNAc-conjugated, inotersen-derived ASO, enhancing liver uptake; similar mechanism of action to inotersen	45 mg subcutaneously every 4 weeks; vitamin A 3000 IU daily supplementation	In ATTRv PN, compared with placebo at 66 weeks: 24.8 point reduction in the progression of neurological impairment (NIS+7 nerve test score) and 19.7 point improvement in QOL (Norfolk QOL-DN score); in ATTR CA: primary endpoint is a composite of cardiovascular mortality and frequency of cardiovascular clinical events at 140 weeks	Minimal ¹⁵³
Patisiran	FDA and EMA approved for ATTRv PN; phase 3 trial completed in ATTR CA—approval declined	Small interfering RNA, targeting the 3' untranslated region of TTR mRNA in the liver; leads to TTR mRNA degradation via cytoplasmic Dicer to form the RNA-RNA-induced silencing complex	0.3 mg/kg intravenous infusion every 3 weeks (max dose 30 mg); vitamin A 3000 IU daily supplementation	In ATTRv PN, compared with placebo at 18 months: 34 point reduction in neurological impairment (NIS+7) and 21.1 point improvement in QOL (Norfolk QOL-DN); in ATTR CA, compared with placebo: 14.7 m improvement in 6-min walk test and 3.7 point improvement in QOL (Kansas City Cardiomyopathy Questionnaire)	Infusion-related reactions (19% of patients) ^{154,155}
Vutrisiran	FDA approved for ATTRv PN and ATTR CA; EMA approved for ATTRv PN	GalNAc-conjugated small interfering RNA targeting TTR mRNA; causes TTR mRNA degradation via RNaseH	25 mg subcutaneously every 12 weeks; vitamin A 3000 IU daily supplementation	In ATTRv PN, compared with placebo at 18 months: 28.6 point reduction in neurological impairment (NIS+7) and 21 point improvement in QOL (Norfolk QOL-DN); in ATTR CA, compared with placebo: 0.72 HR for a composite endpoint of death from any cause and recurrent cardiovascular events	Arthralgias (11% of patients) ^{156,157}
TTR anti-fibril antibody					
ALXN-2200	Phase 3 trial in ATTR CA ongoing (NCT06183931)	Recombinant, IgG1 human monoclonal antibody targeting misfolded, soluble, and insoluble TTRwt and TTRv	Intravenously every 28 days	In ATTRv or ATTRwt CA at 48 months: all-cause mortality and cardiovascular events	Based on a phase 2 study: infusion reactions and arthralgias ¹⁵⁸
Coramitug (PRX004)	Phase 2 trial in ATTR CA completed (NCT05442047)	Human monoclonal antibody targeting misfolded TTR	Intravenously every 28 days	In ATTR CA: change in 6-min walk test and change in NT-proBNP from baseline to week 52	No apparent drug-related serious adverse events ¹⁵⁹
TTR gene editing					
NTLA-2001	Phase 3 trial ongoing in ATTR CA (NCT06128629)	Lipid nanoparticle encapsulated gRNA targeting TTR and Cas9 mRNA; leads to double strand break and non-homologous end joining of the TTR gene, leading to TTR knock out	Single intravenous infusion of NTLA-2001 (55 mg)	In ATTRv or ATTRwt CA at 18–48 months: a composite endpoint of cardiovascular mortality and cardiovascular events	Based on a phase 1 study: vitamin A deficiency ¹⁶⁰
ASO=antisense oligonucleotide. ATTR=transferrin amyloidosis. ATTRv=transferrin amyloidosis variant. ATTRwt=transferrin amyloidosis wild-type. CA=amyloid cardiomyopathy. EMA=European Medicines Agency. FDA=Food and Drug Administration. HR=hazard ratio. IU=international units. NIS+7=Neuropathy Impairment Score+7. NSAID=non-steroidal anti-inflammatory drug. PN=polyneuropathy. QOL=quality of life. QOL-DN=Quality of Life in Diabetic Neuropathy score. SF-36=Short Form-36 Health Survey. TTR=transferrin.					

Table 2: Results for ATTR therapies from completed or ongoing phase 3 clinical studies

Venetoclax is an orally bioavailable inhibitor of the antiapoptotic factor B-cell lymphoma 2 (BCL2) and is approved by the US Food and Drug Administration for chronic lymphocytic leukaemia and acute myeloid leukaemia.^{127,128} Venetoclax showed disproportionate efficacy in patients with multiple myeloma harbouring t(11;14), which serves as a biomarker for increased BCL2 dependency.^{129,130} Venetoclax (monotherapy or in combination) elicits frequent haematological responses in patients with t(11;14) AL amyloidosis, including in the setting of DaraCyBorD refractory disease.^{131–134} These results have led to ongoing prospective studies in patients with relapsed (NCT04847453 and NCT05451771) and newly diagnosed (NCT06192979 and NCT06629818) t(11;14) AL amyloidosis.

B-cell maturation antigen (BCMA)-targeting BiTEs and chimeric antigen receptor T (CAR-T) cells have shown unprecedented efficacy in patients with AL amyloidosis.^{135–138} In small retrospective studies, fixed-duration BiTE teclistimab and elranatamab led to frequent, rapid, and profound suppression of FLC that appears to be sustained long-term off treatment.^{101–103} Prospective studies of BMCA BiTEs are ongoing (NCT06569147, NCT06649695, NCT06292780, and NCT06158854). BCMA CAR-T HBI0101 led to a 100% overall response rate in patients with relapsed AL amyloidosis, although the duration of the response was less than 1 year—a larger, prospective trial is ongoing (NCT06097832).¹³⁹ A phase 1/2 clinical trial of a CD19 and BCMA dual targeting CAR-T (AZD0120) is currently enrolling (NCT07081646).

Anti-fibril mAbs

Successful eradication of AL plasma cells normalises FLC, impeding new amyloid deposition. However, plasma cell-directed therapies have no activity against amyloid deposits in tissues. In preclinical models, the anti-fibril mAbs birtamimab (formerly known as NEOD-001) and anselamimab (formerly known as CAEL-101) bind to amyloid fibrils and recruit macrophages, leading to fibril phagocytosis.^{140–142} Neither agent met the primary endpoint in phase 3 studies evaluating their combination with plasma cell-directed therapies in patients newly diagnosed with AL amyloidosis with advanced cardiomyopathy (NCT04973137, NCT04512235, and NCT04504825). However, anselamimab, generated by immunising mice with the variable region of an amyloidogenic human κ light chain, improved survival and reduced the frequency of cardiovascular-related hospital admissions in patients with κ AL amyloidosis.¹⁴³ Anti-fibril mAbs generated through λ light chain immunisation are in development and will enter clinical trials soon. A first-in-class λ FLC stabiliser, which, in preclinical studies, has been shown to bind to soluble amyloidogenic λ FLCs with avidity, impeding their aggregation, is in early-phase clinical development.^{144,145}

Therapeutic approaches for treating patients with ATTR amyloidosis

Similar to AL amyloidosis, validation of ATTR amyloidosis staging systems in the context of novel therapies is key to ensure these models remain relevant (appendix p 9).¹⁴⁶ ATTR amyloidosis therapies target one of three pathogenetic steps: (1) hepatic TTR production (silencers, gene editing); (2) TTR tetramer instability (stabilisers); and (3) amyloid fibril deposition (depleters; table 2, figure 4). Given the availability of several safe and effective TTR silencers and stabilisers, interest is growing in the use of these agents to prevent cardiomyopathy in patients at risk (ie, ATTRv carriers), outlining the need to better understand the mechanisms underpinning ATTRv clinical penetrance.¹⁶¹

TTR gene silencers

Small interfering RNA therapies (patisiran and vutrisiran) or antisense oligonucleotides (inotersen and eplontersen) effectively reduce hepatic TTR messenger RNA concentration, thus decreasing newly synthesised TTR. These agents bind TTRwt or TTRv RNA transcripts, triggering their degradation via nuclear ribonuclease H (inotersen and eplontersen) or cytoplasmic dicer (patisiran and vutrisiran).

In randomised, placebo-controlled studies in patients with ATTRv with polyneuropathy, both patisiran and inotersen decreased the rate of neuropathy progression as determined by the modified Neuropathy Impairment Score+7.^{152,154} Next-generation small interfering RNA (vutrisiran) and antisense oligonucleotides (eplontersen) showed efficacy, with improved pharmacokinetics and safety profile, particularly for eplontersen, for which thrombocytopenia and glomerulonephritis associated with inotersen were not observed.^{153,156}

Initially approved for the treatment of ATTRv with polyneuropathy, TTR gene silencers were subsequently evaluated in cardiomyopathy.¹⁵⁵ A phase 3 randomised, placebo-controlled trial of vutrisiran in both ATTRv and ATTRwt cardiomyopathy showed a lower risk of death from any cause and of cardiovascular events in the investigational group than did the placebo, leading to its approval for this indication in Europe and the USA.¹⁵⁷ Results from a phase 3 placebo-controlled, randomised trial of eplontersen in both ATTRv and ATTRwt cardiomyopathies are expected in 2026 (NCT04136171).

TTR stabilisers

TTR stabilisers prevent TTR tetramer dissociation into dimeric and, subsequently, monomeric, aggregation-prone forms (figure 4). Tafamidis binds the thyroxine-binding sites of TTR homotetramer and is the first approved TTR stabiliser for patients with ATTRwt or ATTRv cardiomyopathy.¹⁴⁷ Tafamidis led to a 13% absolute reduction in mortality (42.9% vs 29.5%) and reduced cardiovascular and all-cause hospital admissions. Although the efficacy was greater in participants with less

symptomatic heart failure (New York Heart Association classes I and II), a survival advantage for patients with New York Heart Association class III symptoms was also observed. Moreover, tafamidis attenuated the decline in the 6-min walk test and in health-related quality of life compared with placebo, starting at 6 months from treatment initiation. In a randomised trial, tafamidis showed a reduction in the progression of neurological symptoms, suggesting utility in patients with ATTRv polyneuropathy.¹⁴⁸

The TTR stabiliser acoramidis was developed by mimicking the tetramer-stabilising properties of the p.Thr139Met TTR mutation. In a placebo-randomised phase 3 trial in patients with ATTR cardiomyopathy, acoramidis showed superiority according to a hierarchical analysis including all-cause mortality, admission to hospital due to cardiovascular causes, and changes in NT-proBNP and 6-min walk test, with an overall hazard ratio of 1.8 ($p=0.0001$).¹⁴⁹ Acoramidis led to statistically significant reductions in the frequency of cardiovascular-related hospital admissions and to a 6.4% absolute (25% relative) risk reduction in mortality, leading to its approval in the USA and Europe in 2024–25. An early and sustained increase in TTR serum concentration with acoramidis was independently associated with outcome, suggesting its utility as a biomarker of tetramer stabilisation.¹⁵⁰

Highly unstable TTR variants, such as p.Ser72Pro, might benefit from combination therapy with both a TTR stabiliser and a silencer to fully suppress fibrillogenesis. Prospective studies are warranted.

Diflunisal is a non-steroidal anti-inflammatory drug that stabilises the TTR tetramer in vitro. In a placebo-controlled trial in ATTRv with polyneuropathy, diflunisal slowed the progression of Neuropathy Impairment Score+7, and improved the quality of life (Short Form-36 Health Survey questionnaire).¹⁵¹ In a small study of patients with ATTRwt cardiomyopathy, diflunisal improved overall survival and stabilised clinical and echocardiographic markers of disease as compared with placebo.¹⁵¹ However, patients in the investigational group were younger, with lower NT-proBNP and better renal function. The use of diflunisal in ATTR amyloidosis remains off-label, and non-steroidal anti-inflammatory drug-related side-effects, including renal dysfunction, reduce its tolerability in patients who are frail.

Gene editing and liver transplantation

ATTRv amyloidosis is inherited as a monogenic, autosomal dominant disorder, providing an opportunity for curative-intent gene therapy. NTLA-2001 is a lipid nanoparticle containing Cas9 mRNA and a single guide RNA designed to knock out *TTR*. In a phase 1 study, a single infusion of NTLA-2001 led to a dose-dependent, sustained decline in serum TTR in patients with ATTRv amyloidosis with polyneuropathy.¹⁶⁰ A phase 3, double-blind, placebo-controlled trial evaluating the safety and

efficacy of NTLA-2001 in patients with ATTR cardiomyopathy is ongoing (NCT06128629). As TTR is primarily produced by hepatocytes, liver transplantation can be curative for patients with ATTRv amyloidosis.¹⁶² However, pre-existing ATTR amyloidosis-related organ dysfunction is typically irreversible, thus limiting its indication to patients without severe organ (particularly heart) impairment.¹⁶³ Given the availability of highly effective stabiliser and silencing therapies, liver transplantation is no longer considered a standard management approach for ATTR cardiomyopathy.

Anti-fibril mAbs (depleters)

Clinical trials of three distinct anti-fibrillary mAbs (PRX004/NNC6019-0001, AT-02, and N1006/ALXN2220) targeting TTR amyloid are ongoing. In a phase 1 clinical trial, N1006/ALXN2220 markedly reduced surrogate markers of amyloid burden in the myocardium, including NT-proBNP, troponin, and MRI-derived extracellular volume.^{158,159} Intriguingly, naturally occurring, high-titre polyclonal IgG antibodies to human TTR fibrils have been reported in three patients who showed MRI-based evidence of ATTR amyloidosis regression.¹⁶⁴ These data suggest that the immune system can mount a response against TTR amyloid.

Uniform, validated cardiac response criteria are lacking in ATTR amyloidosis, hampering the interpretation and comparison of clinical trials. To this end, interest is growing in the use of imaging biomarkers, such as MRI-based extracellular volume or TTE-based global longitudinal strain, but operator dependency and inconsistent measurement reproducibility limit their broad applicability.

Ancillary care for patients with amyloidosis

A multidisciplinary approach to the care of patients with systemic amyloidosis is essential to maximise quality of life and survival. Early referral to multidisciplinary centres is recommended to discuss diagnosis, therapy, and prognosis. As many resources as needed should be offered to patients and their families, since even highly effective treatments are burdened with side-effects and organ response does not typically occur for months or years. Caregivers and family members play a pivotal role in treating symptoms, administering medications, coordinating visits, and providing emotional and psychological support to patients. Caregiver burnout is a real risk, and every effort should be made to mitigate it.

Management of patients with amyloid cardiomyopathy requires expert cardiology consultation. Decompensated heart failure, typically heart failure with preserved ejection fraction, and arrhythmias are common presenting manifestations. Careful pre-load reduction with a combination of loop diuretics and proximal and distal tubule diuretics is important to avoid admission to hospital and improve quality of life. Atrial tachyarrhythmias are often refractory to cardioversion, requiring

consideration for pacemaker placement. Conduction system disease remains the leading indication for pacemaker implantation in amyloid cardiomyopathy. No clear consensus exists on intracardiac defibrillator device placement for the primary prevention of cardiac arrest in patients with asymptomatic ventricular arrhythmia. Historical data showed no improvement in outcome, as most cardiac deaths in cardiac amyloidosis are secondary to pulseless electrical activity. However, given the remarkable improvement in survival and the increased interest in offering cardiac transplantation, we recommend that an intracardiac defibrillator device be carefully considered for primary prophylaxis in patients with cardiac amyloidosis who appear well poised for long-term survival, based on plasma cell-directed treatment response and suitability for cardiac transplantation.^{165–167} In the absence of major contraindications, anticoagulation should be implemented in patients with atrial tachyarrhythmia and strongly considered in patients without documented atrial tachyarrhythmia who present with poor contractile atrial function. Due to their ease of use, direct oral anticoagulants are often the preferred choice. β blockers are generally poorly tolerated in patients with cardiac amyloidosis, due to their negative inotropic and chronotropic effects, and should be carefully administered only in selected circumstances.¹⁶⁸ Digoxin has been historically contraindicated in patients with amyloid cardiomyopathy due to the risk of overdose secondary to absorption of the drug on amyloid fibrils themselves. However, with careful management and dose monitoring, digoxin can be used in patients with cardiac amyloidosis. Reports suggest that SGLT2 inhibitors are safe and potentially beneficial for patients with amyloid cardiomyopathy.^{169,170}

Nephrotic syndrome can cause secondary hyperlipidaemia and hypothyroidism. Improvement and resolution of these conditions depend on a substantial reduction in proteinuria, which might not occur for years. Therefore, we recommend that hyperlipidaemia and hypothyroidism are treated appropriately and monitored for resolution, with potential drug discontinuation over time. Similarly, nephrotic syndrome leads to a pro-coagulable state, and intermediate or full dose anticoagulation, depending on cumulative risks, should be considered in patients without contraindications, particularly if hypoalbuminaemia of less than 3 g/dL is present.^{171,172} Diuretics are often required to manage volume overload, and recurrent paracentesis might be indicated in patients with ascites. Renal function should be carefully monitored, and routine care, as per chronic kidney disease guidelines, should be implemented as appropriate.¹⁷³

Amyloid-related peripheral neuropathy can be debilitating and often painful. Appropriate management with medications such as gabapentin, pregabalin, or duloxetine is key to alleviate neuropathic pain; we recommend referral to a pain clinic whenever possible. Gastrointestinal side-effects are frequent and diverse. The support of gastroenterologists is crucial for accurate diagnosis, the treatment of reversible causes such as *Helicobacter pylori* infection or peptic ulcer disease, and the optimisation of symptomatic care. A more systematic implementation of tools for longitudinal assessment of patient-reported outcomes and symptom burden is important for gauging treatment intensity, adjusting palliative medications, and focusing on concerns most relevant to each patient.

Finally, solid organ transplantation should be considered in patients with AL amyloidosis with signs or symptoms of progressive organ dysfunction, particularly if young and in a deep and persistent haematological remission.¹⁷⁴ Because highly effective drugs targeting the aetiopathogenesis of AL and ATTR amyloidosis are now available, the eligibility for and appeal of solid organ transplantation are expected to increase in the future. In fact, long-term outcomes of renal and heart transplant in patients with AL amyloidosis are projected to be similar to other disease subgroups.¹⁷⁵ Thus, harmonisation of selection criteria for solid organ transplantation across different transplant centres and amyloidosis programmes is a worthwhile aim.¹⁷⁶

Future directions

Decades of fundamental research have led to discoveries that are changing the lives of patients with AL and ATTR amyloidosis. Basic science will continue to have a prime role in deepening the understanding of these conditions and identifying targetable vulnerabilities (figure 5). Novel therapies, alongside improved supportive measures, refined patient stratification, and increased accessibility to multidisciplinary care, have resulted in continuous improvements in survival and quality of life.

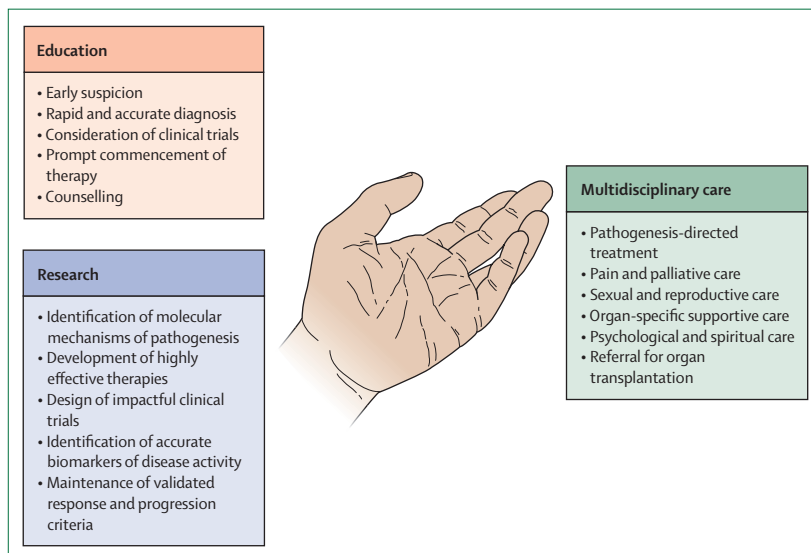


Figure 5: Improving outcomes for patients with amyloidosis

Education, multidisciplinary care, and research are the cornerstones of efforts to improve the lives of patients with amyloidosis.

Nevertheless, the work is not done. First and foremost, diagnostic delays remain frequent, negatively impacting quality of life and increasing the risk for early mortality. The relentless work of non-profit organisations worldwide has been instrumental in raising awareness of amyloidosis in the community, among health-care trainees, and with regulatory agencies. Researchers and clinicians should continue to work on identifying accurate biomarkers of early disease activity, and advances might come from leveraging the chemistry of proteolytic cleavage and the organised deposition of amyloidogenic proteins to develop laboratory and imaging tests for early diagnosis.^{27,45} Addressing the high cost of therapy, especially in ATTR, is key to allow equitable access to approved, effective drugs.

Second, amyloidoses are increasingly recognised as a prevalent cause of symptoms and adverse outcomes, including heart failure and death. As both AL and ATTR amyloidosis are disorders of ageing, their prevalence is expected to increase in high-income countries, particularly as diagnostic tools are applied more consistently worldwide. To this end, research and development will be key, and a transparent, complementary, and productive liaison with pharmaceutical companies could accelerate the translation of laboratory findings to benefit patients.

Third, as novel agents steadily enter clinical development, the broader amyloidosis community should agree on accurate, validated response criteria and surrogate biomarkers for outcome. For example, growing data are being produced in AL amyloidosis regarding the prognostic impact of minimal residual assessment in peripheral blood or bone marrow.^{177–179} Therefore, the validation of these tools as adequate endpoints in prospective studies is essential to accelerate the endorsement of new therapies by national guidelines and regulatory agencies.

Finally, we recognise that patients newly diagnosed with amyloidosis, and their families, embark on a long journey that can include frequent setbacks and severe consequences for their psychological, emotional, physical, and financial wellbeing. Efforts should be made at the regulatory and insurance levels to break down barriers that restrict or obfuscate access to adequate medical care or that delay the approval of novel treatments. These barriers are disproportionately larger for patients from minoritised ethnic communities, reflecting the current social divide.

Conclusions

Over the past decade, remarkable advancements in the treatment of AL and ATTR amyloidosis have been accomplished. We look with optimism at what lies ahead for the future. Commitment to care and investment in research are essential to tackling amyloidosis (figure 5). The overarching goal is to develop screening tests for early detection that can be implemented globally; to discover highly effective, laboratory-based treatments that target the intrinsic vulnerabilities of these diseases

with curative intent; and to improve access to multidisciplinary care across racial and socioeconomic differences.

Contributors

GB: Introduction, Molecular mechanisms underlying amyloidogenesis, Ancillary care for patients with amyloidosis, Conclusion and future directions. GM: Epidemiology, Diagnostic considerations. VS and AW: Comprehensive strategies for treating patients with AL amyloidosis. MG and SAMC: Therapeutic approaches for treating patients with ATTR amyloidosis.

Declaration of interests

GB reports consultancy, honoraria, research funding, or membership on an advisory committee for Pfizer, Janssen Pharmaceuticals, AstraZeneca/Alexion and Prothena Biosciences. VS reports consultancy, honoraria, or research funding from, and membership of an advisory committee for Janssen Pharmaceuticals, Alexion, Prothena Biosciences, Celgene, Takeda Pharmaceuticals, AbbVie, Regeneron, Pfizer, Nexcella, and AstraZeneca. MG reports research and consulting fees from, and sits on advisory boards or steering committees of, Alnylam Pharmaceuticals, AstraZeneca, Attralus, BridgeBio, Intellia Therapeutics, Janssen Pharmaceuticals, Novo Nordisk, and Pfizer, with all compensation going to the Mayo Foundation for Education and Research, and no personal compensation. AW reports honoraria and consulting fees from, and sits on advisory boards for, Alexion, Prothena Biosciences, Janssen Pharmaceuticals, Pfizer, Regeneron, and GlaxoSmithKline. SAMC reports consulting and speaker fees from, and sits on the advisory board for, Pfizer, Alexion, AstraZeneca, Bayer, BridgeBio, Alnylam Pharmaceuticals, Intellia Therapeutics, Novo Nordisk, and Life Molecular Imaging. SAMC also reports site research support from Alnylam Pharmaceuticals, Ionis Pharmaceuticals, AstraZeneca, Alexion, and Intellia Therapeutics. GM declares no competing interests.

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References

- Merlini G, Dispenzieri A, Sanchoawala V, et al. Systemic immunoglobulin light chain amyloidosis. *Nat Rev Dis Primers* 2018; 4: 38.
- Kyle RA. Amyloidosis: a convoluted story. *Br J Haematol* 2001; 114: 529–38.
- Virchow R. On the course of the amyloid degeneration. *J Cell Sci* 1856; s1–4: 135–42.
- Cohen AS, Calkins E. Electron microscopic observations on a fibrous component in amyloid of diverse origins. *Nature* 1959; 183: 1202–03.
- Glenner GG, Terry W, Harada M, Isersky C, Page D. Amyloid fibril proteins: proof of homology with immunoglobulin light chains by sequence analyses. *Science* 1971; 172: 1150–51.
- Buxbaum JN, Dispenzieri A, Eisenberg DS, et al. Amyloid nomenclature 2022: update, novel proteins, and recommendations by the International Society of Amyloidosis (ISA) Nomenclature Committee. *Amyloid* 2022; 29: 213–19.
- Buxbaum JN, Eisenberg DS, Fändrich M, et al. Amyloid nomenclature 2024: update, novel proteins, and recommendations by the International Society of Amyloidosis (ISA) Nomenclature Committee. *Amyloid* 2024; 31: 249–56.
- Rapezzi C, Riva L, Quarta CC, et al. Gender-related risk of myocardial involvement in systemic amyloidosis. *Amyloid* 2008; 15: 40–48.
- González-López E, Gallego-Delgado M, Guzzo-Merello G, et al. Wild-type transthyretin amyloidosis as a cause of heart failure with preserved ejection fraction. *Eur Heart J* 2015; 36: 2585–94.

- 10 Patel RK, Ioannou A, Razvi Y, et al. Sex differences among patients with transthyretin amyloid cardiomyopathy—from diagnosis to prognosis. *Eur J Heart Fail* 2022; **24**: 2355–63.
- 11 Mohammed SF, Mirzoyev SA, Edwards WD, et al. Left ventricular amyloid deposition in patients with heart failure and preserved ejection fraction. *JACC Heart Fail* 2014; **2**: 113–22.
- 12 Grogan M, Scott CG, Kyle RA, et al. Natural history of wild-type transthyretin cardiac amyloidosis and risk stratification using a novel staging system. *J Am Coll Cardiol* 2016; **68**: 1014–20.
- 13 Ream S, Ma J, Rodríguez T, et al. Ethnic/racial differences in risk factors and clinical outcomes among patients with amyloidosis. *Am J Med Sci* 2023; **365**: 232–41.
- 14 Staron A, Connors LH, Zheng L, Doros G, Sanchorawala V. Race/ethnicity in systemic AL amyloidosis: perspectives on disease and outcome disparities. *Blood Cancer J* 2020; **10**: 118.
- 15 Ruberg FL, Teruya S, Helmke S, et al. Transthyretin cardiac amyloidosis in older Black and Hispanic individuals with heart failure. *JAMA Cardiol* 2025; **10**: 1034–43.
- 16 Kyle RA, Larson DR, Kurtin PJ, et al. Incidence of AL amyloidosis in Olmsted County, Minnesota, 1990 through 2015. *Mayo Clin Proc* 2019; **94**: 465–71.
- 17 Buxbaum JN, Ruberg FL. Transthyretin V122I (pV142I)* cardiac amyloidosis: an age-dependent autosomal dominant cardiomyopathy too common to be overlooked as a cause of significant heart disease in elderly African Americans. *Genet Med* 2017; **19**: 733–42.
- 18 Hahn VS, Yanek LR, Vaishnav J, et al. Endomyocardial biopsy characterization of heart failure with preserved ejection fraction and prevalence of cardiac amyloidosis. *JACC Heart Fail* 2020; **8**: 712–24.
- 19 AbouEzzeddine OF, Davies DR, Scott CG, et al. Prevalence of transthyretin amyloid cardiomyopathy in heart failure with preserved ejection fraction. *JAMA Cardiol* 2021; **6**: 1267–74.
- 20 Westin O, Fosbol EL, Maurer MS, et al. Screening for cardiac amyloidosis 5 to 15 years after surgery for bilateral carpal tunnel syndrome. *J Am Coll Cardiol* 2022; **80**: 967–77.
- 21 Madhani A, Sabogal N, Massillon D, et al. Clinical penetrance of the transthyretin V122I variant in older Black patients with heart failure: the SCAN-MP (Screening for Cardiac Amyloidosis With Nuclear Imaging in Minority Populations) study. *J Am Heart Assoc* 2023; **12**: e028973.
- 22 Gillmore JD, Maurer MS, Falk RH, et al. Nonbiopsy diagnosis of cardiac transthyretin amyloidosis. *Circulation* 2016; **133**: 2404–12.
- 23 Weiss BM, Hebreo J, Cordaro DV, et al. Increased serum free light chains precede the presentation of immunoglobulin light chain amyloidosis. *J Clin Oncol* 2014; **32**: 2699–704.
- 24 Merlini G. AL amyloidosis: from molecular mechanisms to targeted therapies. *Hematology Am Soc Hematol Educ Program* 2017; **2017**: 1–12.
- 25 Palladini G, Basset M, Milani P, et al. Biomarker-based screening of organ dysfunction in patients with MGUS allows early diagnosis of AL amyloidosis. *Blood* 2017; **130** (suppl 1): 1760.
- 26 Mangiacavalli S, Milani P, Cartia CS, et al. Feasibility of a biomarker-based screening for pre-symptomatic AL amyloidosis in patients with intermediate/high-risk MGUS. *Am J Hematol* 2025; **100**: 342–45.
- 27 Jiang X, Wang J, Li YC, et al. Blood-based diagnostic assay for λ light chain amyloidosis through quantification of an amyloidogenicity-indicating neo-epitope. *Blood* 2023; **142** (suppl 1): 536.
- 28 Zhou P, Kugelmass A, Toskic D, et al. Seeking AL amyloidosis very early: the SAVE trial—identifying clonal lambda light chain genes in patients with MGUS or smoldering multiple myeloma. *Blood* 2018; **132** (suppl 1): 1903.
- 29 Garofalo M, Piccoli L, Romeo M, et al. Machine learning analyses of antibody somatic mutations predict immunoglobulin light chain toxicity. *Nat Commun* 2021; **12**: 3532.
- 30 Milandri A, Farioli A, Gagliardi C, et al. Carpal tunnel syndrome in cardiac amyloidosis: implications for early diagnosis and prognostic role across the spectrum of aetiologies. *Eur J Heart Fail* 2020; **22**: 507–15.
- 31 Cipriani A, De Michieli L, Porcari A, et al. Low QRS voltages in cardiac amyloidosis: clinical correlates and prognostic value. *JACC Cardiooncol* 2022; **4**: 458–70.
- 32 Goto S, Mahara K, Beussink-Nelson L, et al. Artificial intelligence-enabled fully automated detection of cardiac amyloidosis using electrocardiograms and echocardiograms. *Nat Commun* 2021; **12**: 2726.
- 33 Harmon DM, Mangold K, Suarez AB, et al. Postdevelopment performance and validation of the artificial intelligence-enhanced electrocardiogram for detection of cardiac amyloidosis. *JACC Adv* 2023; **2**: 100612.
- 34 Schrutka L, Anner P, Agibetov A, et al. Machine learning-derived electrocardiographic algorithm for the detection of cardiac amyloidosis. *Heart* 2022; **108**: 1137–47.
- 35 Dorbala S, Ando Y, Bokhari S, et al. ASNC/AHA/ASE/EANM/HFSA/ISA/SCMR/SNMMI expert consensus recommendations for multimodality imaging in cardiac amyloidosis: part 1 of 2-evidence base and standardized methods of imaging. *Circ Cardiovasc Imaging* 2021; **14**: e000029.
- 36 Khan LA, Shaikh FH, Khan MS, et al. Artificial intelligence-enhanced electrocardiogram for the diagnosis of cardiac amyloidosis: a systemic review and meta-analysis. *Curr Probl Cardiol* 2024; **49**: 102860.
- 37 Dorbala S, Cuddy S, Falk RH. How to image cardiac amyloidosis: a practical approach. *JACC Cardiovasc Imaging* 2020; **13**: 1368–83.
- 38 Ioannou A, Patel RK, Razvi Y, et al. Multi-imaging characterization of cardiac phenotype in different types of amyloidosis. *JACC Cardiovasc Imaging* 2023; **16**: 464–77.
- 39 Maurer MS, Bokhari S, Damy T, et al. Expert Consensus Recommendations for the suspicion and diagnosis of transthyretin cardiac amyloidosis. *Circ Heart Fail* 2019; **12**: e006075.
- 40 Geller HI, Singh A, Mirto TM, et al. Prevalence of monoclonal gammopathy in wild-type transthyretin amyloidosis. *Mayo Clin Proc* 2017; **92**: 1800–05.
- 41 Gami A, Woller J, Scheel P, et al. Coexistence of light chain and transthyretin cardiac amyloidosis. *JACC Case Rep* 2024; **29**: 102285.
- 42 Azevedo Coutinho MC, Cortez-Dias N, Cantinho G, et al. The sensitivity of DPD scintigraphy to detect transthyretin cardiac amyloidosis in V30M mutation depends on the phenotypic expression of the disease. *Amyloid* 2020; **27**: 174–83.
- 43 Redzevi B, Kamani CH, Maurizi N, Theaudin M, Prior J, Monney P. False negative ^{99m}Tc -DPD scintigraphy in pVal50Met (Val30Met) hereditary transthyretin amyloidosis. *CJC Open* 2024; **6**: 118–21.
- 44 Genovesi D, Vergaro G, Giorgetti A, et al. [^{18}F]-Florbetaben PET/CT for differential diagnosis among cardiac immunoglobulin light chain, transthyretin amyloidosis, and mimicking conditions. *JACC Cardiovasc Imaging* 2021; **14**: 246–55.
- 45 Wall JS, Martin EB, Lands R, et al. Cardiac amyloid detection by PET/CT imaging of iodine (^{124}I) evuzamitide (^{124}I -p5+14): a phase 1/2 study. *JACC Cardiovasc Imaging* 2023; **16**: 1433–48.
- 46 Clerc OF, Romero Pabón AR, Cuddy SAM, et al. Assessment of cardiac involvement in rare forms of amyloidosis using PET/CT with ^{124}I -evuzamitide (^{124}I -p5+14, AT-01). *J Nucl Cardiol* 2026; published online April 2. <https://doi.org/10.1016/j.nuclcard.2026.102707>.
- 47 Dorbala S, Kijewski MF. Molecular imaging of systemic and cardiac amyloidosis: recent advances and focus on the future. *J Nucl Med* 2023; **64** (suppl 2): 20–28s.
- 48 Dasari S, Theis JD, Vrana JA, et al. Amyloid typing by mass spectrometry in clinical practice: a comprehensive review of 16,175 samples. *Mayo Clin Proc* 2020; **95**: 1852–64.
- 49 Naiki H, Johnson M, Walters K, et al. Global patterns of amyloid typing: results of a survey by the International Society of Amyloidosis (ISA). *Amyloid* 2025; **32**: 139–44.
- 50 Hill MM, Dasari S, Mollee P, et al. The clinical impact of proteomics in amyloid typing. *Mayo Clin Proc* 2021; **96**: 1122–27.
- 51 Schönland SO, Hegenbart U, Bochtler T, et al. Immunohistochemistry in the classification of systemic forms of amyloidosis: a systematic investigation of 117 patients. *Blood* 2012; **119**: 488–93.
- 52 Gonzalez-Lopez E, McPhail ED, Salas-Anton C, et al. Histological typing in patients with cardiac amyloidosis: JACC review topic of the week. *J Am Coll Cardiol* 2024; **83**: 1085–99.
- 53 Riek R. The three-dimensional structures of amyloids. *Cold Spring Harb Perspect Biol* 2017; **9**: a023572.
- 54 Goldschmidt L, Teng PK, Riek R, Eisenberg D. Identifying the amyloids, proteins capable of forming amyloid-like fibrils. *Proc Natl Acad Sci USA* 2010; **107**: 3487–92.
- 55 Quintas A, Saraiva MJ, Brito RM. The tetrameric protein transthyretin dissociates to a non-native monomer in solution. A novel model for amyloidogenesis. *J Biol Chem* 1999; **274**: 32943–49.

- 56 Booth DR, Tan SY, Booth SE, et al. Hereditary hepatic and systemic amyloidosis caused by a new deletion/insertion mutation in the apolipoprotein AI gene. *J Clin Invest* 1996; **97**: 2714–21.
- 57 Gregorini G, Izzi C, Ravani P, et al. Tubulointerstitial nephritis is a dominant feature of hereditary apolipoprotein A-I amyloidosis. *Kidney Int* 2015; **87**: 1223–29.
- 58 Planté-Bordeneuve V, Carayol J, Ferreira A, et al. Genetic study of transthyretin amyloid neuropathies: carrier risks among French and Portuguese families. *J Med Genet* 2003; **40**: e120.
- 59 Rapezzi C, Arbustini E, Caforio ALP, et al. Diagnostic work-up in cardiomyopathies: bridging the gap between clinical phenotypes and final diagnosis. A position statement from the ESC Working Group on Myocardial and Pericardial Diseases. *Eur Heart J* 2013; **34**: 1448–58.
- 60 Kourelis TV, Dasari S, Theis JD, et al. Clarifying immunoglobulin gene usage in systemic and localized immunoglobulin light-chain amyloidosis by mass spectrometry. *Blood* 2017; **129**: 299–306.
- 61 Kimura K, Tsukamoto S, Miyazaki K, et al. Identification of clonal immunoglobulin λ light-chain gene rearrangements in AL amyloidosis using next-generation sequencing. *Exp Hematol* 2021; **101–02**: 34–41.
- 62 Rademaker L, Karimi-Farsijani S, Andreotti G, et al. Role of mutations and post-translational modifications in systemic AL amyloidosis studied by cryo-EM. *Nat Commun* 2021; **12**: 6434.
- 63 Kumar S, Murray D, Dasari S, et al. Assay to rapidly screen for immunoglobulin light chain glycosylation: a potential path to earlier AL diagnosis for a subset of patients. *Leukemia* 2019; **33**: 254–57.
- 64 Westermark GT, Fändrich M, Lundmark K, Westermark P. Noncerebral amyloidoses: aspects on seeding, cross-seeding, and transmission. *Cold Spring Harb Perspect Med* 2018; **8**: a024323.
- 65 Simons JP, Al-Shawi R, Ellmerich S, et al. Pathogenetic mechanisms of amyloid A amyloidosis. *Proc Natl Acad Sci USA* 2013; **110**: 16115–20.
- 66 Connors LH, Shirahama T, Skinner M, Fenves A, Cohen AS. In vitro formation of amyloid fibrils from intact β_2 -microglobulin. *Biochem Biophys Res Commun* 1985; **131**: 1063–68.
- 67 Niwa T, Katsuzaki T, Momoi T, et al. Modification of β_2 m with advanced glycation end products as observed in dialysis-related amyloidosis by 3-DG accumulating in uremic serum. *Kidney Int* 1996; **49**: 861–67.
- 68 Gomes CM, Vendruscolo M. Systems-level organization of extracellular proteostasis. *Science* 2026; **391**: 1332–38.
- 69 Greene MJ, Sam F, Soo Hoo PT, Patel RS, Seldin DC, Connors LH. Evidence for a functional role of the molecular chaperone clusterin in amyloidotic cardiomyopathy. *Am J Pathol* 2011; **178**: 61–68.
- 70 Magalhães J, Saraiva MJ. Clusterin overexpression and its possible protective role in transthyretin deposition in familial amyloidotic polyneuropathy. *J Neuropathol Exp Neurol* 2011; **70**: 1097–106.
- 71 West J, Satapathy S, Whiten DR, et al. Neuroserpin and transthyretin are extracellular chaperones that preferentially inhibit amyloid formation. *Sci Adv* 2021; **7**: eabf7606.
- 72 Wiseman RL, Mesgarzadeh JS, Hendershot LM. Reshaping endoplasmic reticulum quality control through the unfolded protein response. *Mol Cell* 2022; **82**: 1477–91.
- 73 Hipp MS, Kasturi P, Hartl FU. The proteostasis network and its decline in ageing. *Nat Rev Mol Cell Biol* 2019; **20**: 421–35.
- 74 Walter P, Ron D. The unfolded protein response: from stress pathway to homeostatic regulation. *Science* 2011; **334**: 1081–86.
- 75 Cooley CB, Ryno LM, Plate L, et al. Unfolded protein response activation reduces secretion and extracellular aggregation of amyloidogenic immunoglobulin light chain. *Proc Natl Acad Sci USA* 2014; **111**: 13046–51.
- 76 Plate L, Rius B, Nguyen B, Genereux JC, Kelly JW, Wiseman RL. Quantitative interactome proteomics reveals a molecular basis for ATF6-dependent regulation of a destabilized amyloidogenic protein. *Cell Chem Biol* 2019; **26**: 913–25.
- 77 Charalampous C, Dasari S, McPhail E, et al. A proteomic atlas of kidney amyloidosis provides insights into disease pathogenesis. *Kidney Int* 2024; **105**: 484–95.
- 78 Martin DJ, Ramirez-Alvarado M. Glycosaminoglycans promote fibril formation by amyloidogenic immunoglobulin light chains through a transient interaction. *Biophys Chem* 2011; **158**: 81–89.
- 79 McLaughlin RW, De Stigter JK, Sikkink LA, Baden EM, Ramirez-Alvarado M. The effects of sodium sulfate, glycosaminoglycans, and Congo red on the structure, stability, and amyloid formation of an immunoglobulin light-chain protein. *Protein Sci* 2006; **15**: 1710–22.
- 80 Jackson JW, Foster JS, Martin EB, et al. Collagen inhibits phagocytosis of amyloid in vitro and in vivo and may act as a ‘don’t eat me’ signal. *Amyloid* 2023; **30**: 249–60.
- 81 Kampmann M. Molecular and cellular mechanisms of selective vulnerability in neurodegenerative diseases. *Nat Rev Neurosci* 2024; **25**: 351–71.
- 82 Martinez-Rivas G, Bender S, Sirac C. Understanding AL amyloidosis with a little help from in vivo models. *Front Immunol* 2022; **13**: 1008449.
- 83 Richey T, Foster JS, Williams AD, et al. Macrophage-mediated phagocytosis and dissolution of amyloid-like fibrils in mice, monitored by optical imaging. *Am J Pathol* 2019; **189**: 989–98.
- 84 Suenaga G, Ikeda T, Komohara Y, et al. Involvement of macrophages in the pathogenesis of familial amyloid polyneuropathy and efficacy of human iPS cell-derived macrophages in its treatment. *PLoS One* 2016; **11**: e0163944.
- 85 Misumi Y, Ando Y, Gonçalves NP, Saraiva MJ. Fibroblasts endocytose and degrade transthyretin aggregates in transthyretin-related amyloidosis. *Lab Invest* 2013; **93**: 911–20.
- 86 Nishikage S, Fujisawa A, Endoh H, et al. Amyloid deposition through endocytosis in vascular endothelial cells. *Exp Hematol* 2024; **129**: 104129.
- 87 Azevedo EPC, Guimarães-Costa AB, Torezani GS, et al. Amyloid fibrils trigger the release of neutrophil extracellular traps (NETs), causing fibril fragmentation by NET-associated elastase. *J Biol Chem* 2012; **287**: 37206–18.
- 88 Dittloff KT, Iezzi A, Zhong JX, Mohindra P, Desai TA, Russell B. Transthyretin amyloid fibrils alter primary fibroblast structure, function, and inflammatory gene expression. *Am J Physiol Heart Circ Physiol* 2021; **321**: h149–60.
- 89 Leng F, Edison P. Neuroinflammation and microglial activation in Alzheimer disease: where do we go from here? *Nat Rev Neurol* 2021; **17**: 157–72.
- 90 Wang X, Guo Y, Gao Y, et al. Feasibility of ^{68}Ga -labeled fibroblast activation protein inhibitor PET/CT in light-chain cardiac amyloidosis. *JACC Cardiovasc Imaging* 2022; **15**: 1960–70.
- 91 Dispenzieri A, Gertz MA, Kyle RA, et al. Serum cardiac troponins and N-terminal pro-brain natriuretic peptide: a staging system for primary systemic amyloidosis. *J Clin Oncol* 2004; **22**: 3751–57.
- 92 Kumar S, Dispenzieri A, Lacy MQ, et al. Revised prognostic staging system for light chain amyloidosis incorporating cardiac biomarkers and serum free light chain measurements. *J Clin Oncol* 2012; **30**: 989–95.
- 93 Wechalekar AD, Schonland SO, Kastritis E, et al. A European collaborative study of treatment outcomes in 346 patients with cardiac stage III AL amyloidosis. *Blood* 2013; **121**: 3420–27.
- 94 Khwaja J, Kirkwood AA, Milani P, et al. New validated staging system for light chain (AL) amyloidosis with stage IIIC defining ultra-poor risk: AL International Staging System. *J Clin Oncol* 2026; **44**: 311–20.
- 95 Saunders B, Theodorakou F, Fotiou D, et al. Predictive value of free light chain burden in patients with AL amyloidosis treated with bortezomib-based regimens. *Blood Adv* 2025; **9**: 3771–79.
- 96 Sancharawala V. Systemic light chain amyloidosis. *N Engl J Med* 2024; **390**: 2295–307.
- 97 Kastritis E, Palladini G, Minnema MC, et al, and the ANDROMEDA Trial Investigators. Daratumumab-based treatment for immunoglobulin light-chain amyloidosis. *N Engl J Med* 2021; **385**: 46–58.
- 98 Sancharawala V, Boccadoro M, Gertz M, et al. Guidelines for high dose chemotherapy and stem cell transplantation for systemic AL amyloidosis: EHA-ISA working group guidelines. *Amyloid* 2022; **29**: 1–7.
- 99 Wechalekar AD, Cibeira MT, Gibbs SD, et al. Guidelines for non-transplant chemotherapy for treatment of systemic AL amyloidosis: EHA-ISA working group. *Amyloid* 2023; **30**: 3–17.
- 100 Gustine JN, Staron A, Szalat RE, et al. Predictors of hematologic response and survival with stem cell transplantation in AL amyloidosis: a 25-year longitudinal study. *Am J Hematol* 2022; **97**: 1189–99.

- 101 Vianna P, Chakraborty R, Hossain S, et al. Safety and efficacy of elranatamab in patients with relapsed and/or refractory immunoglobulin light-chain amyloidosis. *Blood* 2025; **146**: 1929–35.
- 102 Chakraborty R, Bhutani D, Maurer MS, Mohan M, Lentzsch S, D'Souza A. Safety and efficacy of teclistamab in systemic immunoglobulin light chain amyloidosis. *Blood Cancer J* 2023; **13**: 172.
- 103 Forgeard N, Elessa D, Carpinteiro A, et al. Teclistamab in relapsed or refractory AL amyloidosis: a multinational retrospective case series. *Blood* 2024; **143**: 734–37.
- 104 Hagen P, D'Souza A. Autologous stem cell transplantation in AL amyloidosis: muddy waters. *Blood Rev* 2024; **68**: 101228.
- 105 Bianchi G, Zhang Y, Comenzo RL. AL amyloidosis: current chemotherapy and immune therapy treatment strategies: JACC: CardioOncology state-of-the-art review. *JACC Cardiooncol* 2021; **3**: 467–87.
- 106 Chakraborty R, Rosenbaum C, Kaur G, et al. First report of outcomes in patients with stage IIb AL amyloidosis treated with dara-VCD front-line therapy. *Br J Haematol* 2023; **201**: 913–16.
- 107 Kastritis E, Minnema MC, Dimopoulos MA, et al. Efficacy and safety of daratumumab monotherapy in newly diagnosed patients with stage 3b light-chain amyloidosis: a phase 2 study by the European Myeloma Network. *Blood* 2023; **142** (suppl 1): 539.
- 108 Oubari S, Hegenbart U, Schoder R, et al. Daratumumab in first-line treatment of patients with light chain amyloidosis and Mayo stage IIb improves treatment response and overall survival. *Haematologica* 2024; **109**: 220–30.
- 109 Palladini G, Milani P, Folli A, et al. Presentation and outcome with second-line treatment in AL amyloidosis previously sensitive to nontransplant therapies. *Blood* 2018; **131**: 525–32.
- 110 Adams J. The development of proteasome inhibitors as anticancer drugs. *Cancer Cell* 2004; **5**: 417–21.
- 111 Reece DE, Sanchorawala V, Hegenbart U, et al, and the VELCADE CAN2007 Study Group. Weekly and twice-weekly bortezomib in patients with systemic AL amyloidosis: results of a phase 1 dose-escalation study. *Blood* 2009; **114**: 1489–97.
- 112 Sanchorawala V, Palladini G, Kukreti V, et al. A phase 1/2 study of the oral proteasome inhibitor ixazomib in relapsed or refractory AL amyloidosis. *Blood* 2017; **130**: 597–605.
- 113 Dispenzieri A, Kastritis E, Wechalekar AD, et al. A randomized phase 3 study of ixazomib-dexamethasone versus physician's choice in relapsed or refractory AL amyloidosis. *Leukemia* 2022; **36**: 225–35.
- 114 Moscvin M, Liacos CI, Chen T, et al. Mutations in the alternative complement pathway in multiple myeloma patients with carfilzomib-induced thrombotic microangiopathy. *Blood Cancer J* 2023; **13**: 31.
- 115 Manwani R, Mahmood S, Sachchithanatham S, et al. Carfilzomib is an effective upfront treatment in AL amyloidosis patients with peripheral and autonomic neuropathy. *Br J Haematol* 2019; **187**: 638–41.
- 116 Arnall JR, Usmani SZ, Adamu H, Mishkin J, Bhutani M. Daratumumab, pomalidomide, and dexamethasone as a bridging therapy to autologous stem cell transplantation in a case of systemic light-chain amyloidosis with advanced cardiac involvement. *J Oncol Pharm Pract* 2018; **25**: 1021–25.
- 117 Sanchorawala V, Sarosiek S, Schulman A, et al. Safety, tolerability, and response rates of daratumumab in relapsed AL amyloidosis: results of a phase 2 study. *Blood* 2020; **135**: 1541–47.
- 118 Roussel M, Merlini G, Chevret S, et al. A prospective phase 2 trial of daratumumab in patients with previously treated systemic light-chain amyloidosis. *Blood* 2020; **135**: 1531–40.
- 119 Parker TL, Rosenthal A, Sanchorawala V, et al. A phase II study of isatuximab (SAR650984) (NSC-795145) for patients with previously treated AL amyloidosis (SWOG S1702; NCT03499808). *Blood* 2020; **136** (suppl 1): 20–21.
- 120 Vohra SR, Ravichandran S, Foard D, et al. A retrospective study of isatuximab-pomalidomidedexamethasone in relapsed/refractory systemic AL amyloidosis. *Haematologica* 2025; **110**: 3109–13.
- 121 Sanchorawala V, Doros G, Shelton AC. Long term outcome of patients treated on clinical trials of immunomodulatory agents for the treatment of immunoglobulin light chain (AL) amyloidosis: a pooled analysis. *Am J Hematol* 2019; **94**: e194–96.
- 122 Tapan U, Seldin DC, Finn KT, et al. Increases in B-type natriuretic peptide (BNP) during treatment with lenalidomide in AL amyloidosis. *Blood* 2010; **116**: 5071–72.
- 123 Kastritis E, Dialoupi I, Gavriatopoulou M, et al. Primary treatment of light-chain amyloidosis with bortezomib, lenalidomide, and dexamethasone. *Blood Adv* 2019; **3**: 3002–09.
- 124 Kastritis E, Leleu X, Arnulf B, et al. Bortezomib, melphalan, and dexamethasone for light-chain amyloidosis. *J Clin Oncol* 2020; **38**: 3252–60.
- 125 Sanchorawala V, Seldin DC, Berk JL, Sloan JM, Doros G, Skinner M. Oral cyclic melphalan and dexamethasone for patients with AL amyloidosis. *Clin Lymphoma Myeloma Leuk* 2010; **10**: 469–72.
- 126 Lentzsch S, Lagos GG, Comenzo RL, et al. Bendamustine with dexamethasone in relapsed/refractory systemic light-chain amyloidosis: results of a phase II study. *J Clin Oncol* 2020; **38**: 1455–62.
- 127 DiNardo CD, Lachowicz CA, Takahashi K, et al. Venetoclax combined with FLAG-IDA induction and consolidation in newly diagnosed acute myeloid leukemia. *Am J Hematol* 2022; **97**: 1035–43.
- 128 Fischer K, Al-Sawaf O, Bahlo J, et al. Venetoclax and obinutuzumab in patients with CLL and coexisting conditions. *N Engl J Med* 2019; **380**: 2225–36.
- 129 Kumar SK, Harrison SJ, Cavo M, et al. Venetoclax or placebo in combination with bortezomib and dexamethasone in patients with relapsed or refractory multiple myeloma (BELLINI): a randomised, double-blind, multicentre, phase 3 trial. *Lancet Oncol* 2020; **21**: 1630–42.
- 130 Morales AA, Kurtoglu M, Matulis SM, et al. Distribution of Bim determines Mcl-1 dependence or codependence with Bcl-xL/Bcl-2 in Mcl-1-expressing myeloma cells. *Blood* 2011; **118**: 1329–39.
- 131 Premkumar VJ, Lentzsch S, Pan S, et al. Venetoclax induces deep hematologic remissions in t(11;14) relapsed/refractory AL amyloidosis. *Blood Cancer J* 2021; **11**: 10.
- 132 Lebel E, Kastritis E, Palladini G, et al. Venetoclax in relapse/refractory AL amyloidosis—a multicenter international retrospective real-world study. *Cancers* 2023; **15**: 1710.
- 133 Orland M, Dima D, Ullah F, et al. Outcomes of venetoclax-based therapy in patients with daratumumab-refractory t(11;14) positive light chain amyloidosis. *Blood* 2023; **142** (suppl 1): 2031.
- 134 Roussel M, Pirotte M, Rizzo O, et al. Venetoclax-targeted therapy in t(11;14) AL amyloidosis patients after frontline daratumumab and CyBorD: a retrospective analysis from the French Amyloidosis Network. *Blood* 2022; **140** (suppl 1): 10086–87.
- 135 Raju N, Berdeja J, Lin Y, et al. Anti-BCMA CAR T-Cell therapy bb2121 in relapsed or refractory multiple myeloma. *N Engl J Med* 2019; **380**: 1726–37.
- 136 Berdeja JG, Madduri D, Usmani SZ, et al. Ciltacabtagene autoleucel, a B-cell maturation antigen-directed chimeric antigen receptor T-cell therapy in patients with relapsed or refractory multiple myeloma (CARTITUDE-1): a phase 1b/2 open-label study. *Lancet* 2021; **398**: 314–24.
- 137 Moreau P, Garfall AL, van de Donk NWCJ, et al. Teclistamab in relapsed or refractory multiple myeloma. *N Engl J Med* 2022; **387**: 495–505.
- 138 Lesokhin AM, Tomasson MH, Arnulf B, et al. Elranatamab in relapsed or refractory multiple myeloma: phase 2 MagnetisMM-3 trial results. *Nat Med* 2023; **29**: 2259–67.
- 139 Lebel E, Asherie N, Kfir-Erenfeld S, et al. Efficacy and safety of anti-B-cell maturation antigen chimeric antigen receptor T-cell for the treatment of relapsed and refractory AL amyloidosis. *J Clin Oncol* 2025; **43**: 2007–16.
- 140 Renz M, Torres R, Dolan PJ, et al. 2A4 binds soluble and insoluble light chain aggregates from AL amyloidosis patients and promotes clearance of amyloid deposits by phagocytosis. *Amyloid* 2016; **23**: 168–77.
- 141 Wall JS, Kennel SJ, Williams A, et al. AL amyloid imaging and therapy with a monoclonal antibody to a cryptic epitope on amyloid fibrils. *PLoS One* 2012; **7**: e52686.
- 142 Edwards CV, Rao N, Bhutani D, et al. Phase 1a/b study of monoclonal antibody CAEL-101 (11–1F4) in patients with AL amyloidosis. *Blood* 2021; **138**: 2632–41.
- 143 Wechalekar AD, Dispenzieri A, Sanchorawala V, et al, and the CARES Investigators. Efficacy and safety of anselamimab in immunoglobulin light chain amyloidosis: results from the randomized CARES trials. *J Clin Oncol* 2026; published online May 29. <https://doi.org/10.1200/JCO-26-00755>.

- 144 Yan NL, Morgan GJ, Petrassi HM, Wilson IA, Kelly JW. Pharmacological stabilization of the native state of full-length immunoglobulin light chains to treat light chain amyloidosis. *Curr Opin Chem Biol* 2023; **75**: 102319.
- 145 Qin B, Jackman AB, Li Y-C, et al. Small molecule kinetic stabilizers reduce amyloidogenicity of free light chain proteins of diverse sequences in λ light chain amyloidosis. *Blood* 2024; **144** (supplement 1): 3372.
- 146 Müller ML, Spethmann S, Messrogli D, et al. Accuracy of established prognostic staging systems for cardiac transthyretin amyloidosis in the tafamidis era. *JACC Adv* 2025; **4**: 101568.
- 147 Maurer MS, Schwartz JH, Gundapaneni B, et al, and the ATTR-ACT Study Investigators. Tafamidis treatment for patients with transthyretin amyloid cardiomyopathy. *N Engl J Med* 2018; **379**: 1007–16.
- 148 Coelho T, Maia LF, Martins da Silva A, et al. Tafamidis for transthyretin familial amyloid polyneuropathy: a randomized, controlled trial. *Neurology* 2012; **79**: 785–92.
- 149 Gillmore JD, Judge DP, Cappelli F, et al, and the ATTRIBUTE-CM Investigators. Efficacy and safety of acoramidis in transthyretin amyloid cardiomyopathy. *N Engl J Med* 2024; **390**: 132–42.
- 150 Maurer MS, Judge DP, Gillmore JD, et al. Early increase in serum transthyretin by acoramidis independently predicts improved survival in TTR amyloid cardiomyopathy. *J Am Coll Cardiol* 2025; **85**: 1911–23.
- 151 Berk JL, Suhr OB, Obici L, et al, and the Diflunisal Trial Consortium. Repurposing diflunisal for familial amyloid polyneuropathy: a randomized clinical trial. *JAMA* 2013; **310**: 2658–67.
- 152 Benson MD, Waddington-Cruz M, Berk JL, et al. Inotersen treatment for patients with hereditary transthyretin amyloidosis. *N Engl J Med* 2018; **379**: 22–31.
- 153 Coelho T, Marques Jr W, Dasgupta NR, et al, and the NEURO-TTRransform Investigators. Eplontersen for hereditary transthyretin amyloidosis with polyneuropathy. *JAMA* 2023; **330**: 1448–58.
- 154 Adams D, Gonzalez-Duarte A, O’Riordan WD, et al. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *N Engl J Med* 2018; **379**: 11–21.
- 155 Maurer MS, Kale P, Fontana M, et al, and the APOLLO-B Trial Investigators. Patisiran treatment in patients with transthyretin cardiac amyloidosis. *N Engl J Med* 2023; **389**: 1553–65.
- 156 Adams D, Tourneval IL, Taylor MS, et al, and the HELIOS-A Collaborators. Efficacy and safety of vutrisiran for patients with hereditary transthyretin-mediated amyloidosis with polyneuropathy: a randomized clinical trial. *Amyloid* 2023; **30**: 18–26.
- 157 Fontana M, Berk JL, Gillmore JD, et al, and the HELIOS-B Trial Investigators. Vutrisiran in patients with transthyretin amyloidosis with cardiomyopathy. *N Engl J Med* 2025; **392**: 33–44.
- 158 Garcia-Pavia P, Aus dem Siepen F, Donal E, et al. Phase 1 trial of antibody NI006 for depletion of cardiac transthyretin amyloid. *N Engl J Med* 2023; **389**: 239–50.
- 159 Fontana M, García-Pavía P, Grogan M, et al. Coramitug, a humanized monoclonal antibody for the treatment of transthyretin amyloid cardiomyopathy: a phase 2, randomized, multicenter, double-blind, placebo-controlled trial. *Circulation* 2026; **153**: 214–25.
- 160 Gillmore JD, Gane E, Taubel J, et al. CRISPR-Cas9 in vivo gene editing for transthyretin amyloidosis. *N Engl J Med* 2021; **385**: 493–502.
- 161 Soman P, Garcia-Pavia P, Gillmore JD, et al. Rationale & design of ACT-EARLY, the Acoramidis Transthyretin Amyloidosis Prevention trial: early detection using cardiac radionuclide imaging. *J Nucl Cardiol* 2024; **38**: 101948.
- 162 Wilczek HE, Tranăng M, Coelho T, Ericzon BG. Amyloidogenic phenotypic variation affects post-transplant outcome of hereditary transthyretin amyloidosis: a retrospective study. *Egastroenterology* 2026; **4**: e100243.
- 163 Carvalho A, Rocha A, Lobato L. Liver transplantation in transthyretin amyloidosis: issues and challenges. *Liver Transpl* 2015; **21**: 282–92.
- 164 Fontana M, Gilbertson J, Verona G, et al. Antibody-associated reversal of ATTR amyloidosis-related cardiomyopathy. *N Engl J Med* 2023; **388**: 2199–201.
- 165 Al-Sadawi M, Henriques MD, Gier C, et al. Effect of defibrillator on all-cause mortality in patients with cardiac amyloidosis. *JACC Adv* 2023; **2**: 100561.
- 166 Bukhari S, Khan B. Prevalence of ventricular arrhythmias and role of implantable cardioverter-defibrillator in cardiac amyloidosis. *J Cardiol* 2023; **81**: 429–33.
- 167 Giancaterino S, Urey MA, Darden D, Hsu JC. Management of arrhythmias in cardiac amyloidosis. *JACC Clin Electrophysiol* 2020; **6**: 351–61.
- 168 Kwok CS, Choy CH, Pinney J, et al. Effect of beta-blockade on mortality in patients with cardiac amyloidosis: a systematic review and meta-analysis. *ESC Heart Fail* 2024; **11**: 3901–10.
- 169 Porcari A, Cappelli F, Nitsche C, et al. SGLT2 inhibitor therapy in patients with transthyretin amyloid cardiomyopathy. *J Am Coll Cardiol* 2024; **83**: 2411–22.
- 170 Lang FM, Teruya S, Cuomo M, et al. Safety and efficacy of SGLT2 inhibitors for amyloid light-chain cardiomyopathy. *J Card Fail* 2024; **30**: 1641–46.
- 171 Gholizadeh Ghouloujeh Z, Glasscock RJ, Al Jurdi A, et al. Venous thromboembolism prevention in nephrotic syndrome: the role of aspirin, vitamin K antagonists, and direct oral anticoagulants. *Kidney Int Rep* 2025; **10**: 1335–45.
- 172 Bever KM, Masha LI, Sun F, et al. Risk factors for venous thromboembolism in immunoglobulin light chain amyloidosis. *Haematologica* 2016; **101**: 86–90.
- 173 Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 Clinical Practice Guideline for the evaluation and management of chronic kidney disease. *Kidney Int* 2024; **105**: s117–314.
- 174 Lyle MA, Rosenthal JL, Nativi Nicolau JN, et al. Heart transplantation for cardiac amyloidosis: Mayo Clinic Consensus Statement. *Mayo Clin Proc* 2025; **100**: 1578–605.
- 175 Lyle MA, Farina JMM, Wiedmeier-Nutor E, et al. Amyloidosis and heart transplantation in a new era. *Clin Transplant* 2025; **39**: e70070.
- 176 Havasi A, Heybeli C, Leung N, et al. Outcomes of renal transplantation in patients with AL amyloidosis: an international collaboration through The International Kidney and Monoclonal Gammopathy Research Group.
- 177 Palladini G, Paiva B, Wechalekar A, et al. Minimal residual disease negativity by next-generation flow cytometry is associated with improved organ response in AL amyloidosis. *Blood Cancer J* 2021; **11**: 34.
- 178 Sarosiek S, Varga C, Jacob A, Fulciniti MT, Munshi N, Santhorawala V. Detection of minimal residual disease by next generation sequencing in AL amyloidosis. *Blood Cancer J* 2021; **11**: 117.
- 179 Bomsztyk J, Ravichandran S, Giles HV, et al. Complete responses in AL amyloidosis are unequal: the impact of free light chain mass spectrometry in AL amyloidosis. *Blood* 2024; **143**: 1259–68.

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